Playtime–wellbeing associations are practically null and vary little across video game genres

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Video game research often treats games as a monolithic category, and single-game or single-platform studies leave open whether findings generalise across the medium. In this Registered Report, we used digital trace data from the Open Play dataset to test whether (H1) total playtime is meaningfully associated with mental wellbeing and (H2) this association differs by game genre. The data comprise 222,000 hours of playtime across 4,128 games from Nintendo, Xbox, and Steam for 1,668 UK and US adults, linked to genre metadata and 6,518 biweekly wellbeing assessments (SWEMWBS). Judged against a preregistered threshold for practical significance, neither within-person fluctuations nor habitual differences in total playtime were meaningfully linked to wellbeing. The preregistered joint test found within-person differences between genres under our multi-genre attribution but not under the preregistered single-genre attribution, and no between-person differences. Under an exploratory hierarchical model, the estimated spread of the within-person genre effects implied that fewer than one of the 14 genres was expected to lie more than 0.06 points per hour from the average genre effect, and a multi-membership model estimated the spread less precisely. An exploratory daily-timescale analysis gave the same picture for total playtime and the hierarchical model's genre spread. Genre differences appear to be the wrong place to look for gaming's wellbeing effects, as genre labels are too coarse a proxy for the psychological experiences that plausibly drive them. We therefore regard finer-grained characterisations of play as the more promising avenue for further study.

Keywords: video games, wellbeing, genres, digital trace data, Registered Report
Words: 11644

Introduction

Identifying how different types of games and gameplay affect wellbeing is vital in light of the contrast between popular sentiment, which treats video games as a harm to those who play them in the latest of a long line of technology panics (Orben, 2020), and empirical evidence that playtime in general is not a meaningful predictor of wellbeing, using data from both specific games (Larrieu et al., 2023) and platforms (Ballou, Sewall, et al., 2024; Ballou, Vuorre, et al., 2025). A common objection to these null findings is that the effect of play must depend on which games people play. Multiple mechanisms have been proposed to explain why aggregate playtime is poorly predictive of wellbeing, including displacement of beneficial activities, satisfaction or frustration of basic psychological needs, and selection effects whereby individuals with lower wellbeing play more (Ballou, Hakman, et al., 2025). Displacement aside, these mechanisms operate at the level of game features or player experience rather than at the level of total time spent; playtime might be, at best, a moderator (Ballou, Hakman, et al., 2025). In causal terms, playing is a compound exposure that comes in many versions, so an average association across all of them answers an ambiguous question (Magnusson et al., 2024).

Given the vast heterogeneity between games, our understanding of the psychological processes and outcomes of digital play has been limited by video game effects research's widespread use of either undifferentiated self-report data, or digital trace data focusing on a single game or a handful of games (Ballou, Vuorre, et al., 2025; Brühlmann et al., 2020; Johannes et al., 2021; Larrieu et al., 2023; Vuorre et al., 2022; Vuorre, Ballou, et al., 2024). While understandable given the limited resources and lack of objective data available to independent researchers, this narrow scope has left the field guessing whether identified relationships are unique to the studied titles or whether they apply universally across games. One step forward is therefore to capture comprehensive play behaviour data across the varied games and gaming platforms users regularly engage in, and to differentiate games using structured taxonomies. The most widespread system for classifying games is game genres, albeit not without some controversy (Clarke et al., 2017; Starosta et al., 2024; Vargas-Iglesias, 2020).

Game Genres and Wellbeing

Consistent null findings for the effects of playtime on wellbeing have generated calls for the unpacking of raw

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This study was preregistered as part of a Stage 1 Programmatic Registered Report; preregistration materials are available at https://osf.io/mvngt/. The Open Play dataset (v1.3.0) is openly available on Zenodo at https://zenodo.org/records/23419571. All analysis code, including the multi-membership model validation simulation, is available at https://github.com/Thomhak/open-play-genres.

decontextualised playtime into more informative measures that consider games' varied characteristics and affordances (Ballou, Sewall, et al., 2024). Game genres offer a structured way to differentiate between video games and their corresponding play experiences (Starosta et al., 2024). Genre differences in player behaviour, cognition, and gaming disorder symptoms are often proposed (André et al., 2024; Dobrowolski et al., 2015; Hazel et al., 2022; Raith et al., 2021): for example, action games have been linked to increased visual attention (Palaus et al., 2017), cognitive and attentional control (Anguera et al., 2013; Bavelier & Green, 2019), and working memory (Blacker et al., 2014). Massively multiplayer online role-playing games (MMORPGs), first-person shooters, real-time strategy games (RTS), and Multiplayer Online Battle Arena (MOBA) games have been associated with higher gaming disorder symptoms (Elliott et al., 2012; King et al., 2019; Na et al., 2017; Rehbein et al., 2021). Player experience also differs by genre: MOBA players report less immersion and autonomy and more frustration than players of single-player genres (Johnson et al., 2015), and role-playing games are the genre players most often perceive as beneficial (Hazel et al., 2022). Randomised controlled trials with casual games and exergames induced positive moods and reduced stress (Huang et al., 2017; Russoniello et al., 2009). These findings suggest that the effects of video games are not uniform and that genre differences exist in both gaming behaviour and its impact on wellbeing. This evidence is, however, between-person: the observational studies compare people who favour different genres, and the trials assign people to one genre for a short period, so to our knowledge no study has asked whether a person's wellbeing rises and falls with their own genre playtime over time.

With more than 10,000 new games released each year (Notis, 2021), an accessible classification system is essential for studying and comparing video games at scale. Genres, while defined by convention and neither exclusive nor exhaustive as labels (Clarke et al., 2017; Starosta et al., 2024; Vargas-Iglesias, 2020), remain the most pragmatic means of categorising video games. Psychologists have differentiated games based on genre for decades (Griffiths, 1993; Wolf, 2002), but must negotiate challenges in definition, overlaps between genres, and the continuous evolution of genres as new games are released. The overlaps are not a marginal problem: in metadata databases such as IGDB, most of the games people play carry more than one genre label (see Methods), which complicates any analysis that files each game under a single genre. Genre labels conflate distinct facets of games, such as gameplay style, purpose, and mood (Lee et al., 2014), and factor-analytic decompositions of crowd-sourced game tags produce genre structures that diverge from conventional labels (Li, 2020). Even the cognitive literature that produced the best-known genre-specific claims now treats genre labels as a weak proxy for the mechanics that matter, because games increasingly blend mechanics across genres, as popular titles such as *Skyrim* do (Bavelier & Green, 2019). Genres are therefore best treated not as definitive taxonomical statements but as crude yet practical tools for organising a growing and diversifying library.

There is a lack of consistent standardisation of game genres, leading to inconsistent classification of video game genres across psychological research (Starosta et al., 2024). Starosta et al. (2024) counted 46 distinct taxonomies across the 96 articles they examined; only 50 articles (52%) drew on an existing or combined source for their classification, and 18 (19%) did not specify how their categories were constructed at all. Genre classifications are typically self-reported by participants or assigned by authors, and the same games are habitually classified into different genres across studies. This subjectivity undermines the validity of existing self-report or researcher-ascribed genre taxonomies for the meaningful comparison of results across studies.

Structured, crowd-sourced metadata repositories such as SteamDB, The Games Database, and the Internet Games Database (IGDB) offer a more reliable alternative, involving both game developers and large player samples in collaboratively maintaining genre labels (Lee et al., 2015; Li & Zhang, 2020; Starosta et al., 2024). Within the constraints of genre as a means of categorising games, metadata repositories best accommodate the fluidity of genres over time (Cohen, 1986) and enable comparative research at scale.

The Present Study

In line with prior evidence (Ballou, Sewall, et al., 2024; Ballou, Vuorre, et al., 2025; Larrieu et al., 2023), our first hypothesis predicts the absence of a meaningful relationship between playtime and wellbeing, now with digital trace data across several platforms and thousands of games:

H1. Total cross-platform playtime is not meaningfully associated with fluctuations in general mental wellbeing over a 2-week period (**H1a**, "within-person") or with average wellbeing over the full study period (**H1b**, "between-person").

In practical terms, with H1a we ask whether a participant's wellbeing in a given two-week wave deviates from their own average when their playtime in that wave does, and with H1b whether participants who habitually play more report different average wellbeing than those who play less. Fluctuations in H1 therefore refer to deviations from a person's own average rather than to changes between consecutive waves. What counts as a meaningful association is fixed in advance by the smallest effect size of interest defined in the Methods.

Second, following previous literature (André et al., 2024; Dobrowolski et al., 2015; Elliott et al., 2012; Hazel et al., 2022; Raith et al., 2021; Rehbein et al., 2021), we predict that genres vary in their relation to wellbeing, with some potentially positive and others potentially negative, varying in strength. Put more formally:

H2. Genres differ in how playtime relates to fluctuations in general mental wellbeing over a 2-week period (**H2a**, "within-person") and to average wellbeing over the full study period (**H2b**, "between-person").

With H2a we ask: when a participant plays more than their own average this wave, does it matter what genres that extra playtime is in? With H2b we ask the between-person complement: among people who play more overall, does it matter whether that habitual extra playtime is concentrated in, for example, RPGs versus Shooters versus other genres? By examining heterogeneity across genres (Vuorre, Kay, et al., 2024), we thus test with H2 whether it is justified to generalise video game effects across genres, the generalisability that single-title studies sometimes assume (Yarkoni, 2022).

Beyond the preregistered hypotheses, we report one exploratory extension that was not preregistered. The biweekly window comprises potentially many individual play sessions and a single retrospective wellbeing judgement, so the preregistered tests leave open whether associations operate on shorter timescales. As such, we re-examine both hypotheses at the daily level.

Methods

Data Source and Participants

The analyses reported here are part of a Stage 1 Registered Report (Ballou, Földes, et al., 2024), and the data collected for this study are part of the Open Play dataset (Ballou, Földes, Vuorre, et al., 2025; release v1.3.0, Ballou et al., 2026), a longitudinal dataset that combined multi-platform video game digital trace data with daily and biweekly psychological measures from adult gamers in the UK and US over a three-month period. Qualifying participants completed the intake survey between Sep 2024 and Jun 2025. The digital trace data span the following periods: Nintendo (May 2022 to Oct 2025), Xbox (May 2022 to Sep 2025), Steam (Nov 2024 to Oct 2025), iOS (Mar 2025 to Aug 2025), Android (Mar 2025 to Aug 2025).

This study received ethical approval from the Social Sciences and Humanities Inter-Divisional Research Ethics Committee at the University of Oxford (OII_CIA_23_107). All participants provided informed consent at the start of the study, including consent to their data being shared openly for reanalysis.

Participants were paid at an average rate of £12/hour, equating to £0.20 for a 1-minute screening, £2 for the 10-minute intake survey (plus £5 for linking at least one account with recent data), £0.80 for each 4-minute daily survey, and £2 for each 10-minute biweekly survey. Participants received a £10 bonus payment for completing at least 24 out of 30 daily surveys and/or 5 out of 6 biweekly surveys.

Of the 32,684 participants who completed the intake survey, 2,310 met the preregistered screening and linking criteria. The analytic sample comprises the 1,668 qualified participants who provided valid SWEMWBS responses in at least one biweekly survey wave. We analysed release v1.3.0 of the dataset (Ballou et al., 2026). We measured playtime for these participants from their linked Nintendo, Xbox, and Steam accounts. Rather than excluding participants and waves without recorded play on these platforms, we retained them with zero hours: 125 participants (7%) had no recorded play in any wave, and 988 of the 6,518 person-waves (15%) had none in that wave. In the full analytic sample, 1,532 participants (92%) had at least one session of a game that we could match to a genre. All 1,668 analytic-sample participants have an age within the 18-40 screening range (967 US, 701 UK; mean age 27 years, 65% men), so the demographic sensitivity analysis starts from the full analytic sample. We report full demographic characteristics, including country-level breakdowns, in Table A0.1.

Measures

Figure 1 shows the distributions and structure of the key variables.

Wellbeing. We assessed wellbeing with the Short Warwick-Edinburgh Mental Well-being Scale (SWEMWBS) (Tennant et al., 2007), a 7-item short form of the WEMWBS covering psychological functioning and subjective wellbeing over the past 2 weeks, with responses on a 5-point Likert scale ranging from “None of the time” to “All of the time.” We score the scale as the arithmetic mean of the 7 items (range: 1–5). Internal consistency across all analytic person-waves was high (McDonald’s ω_t = 0.89, Cronbach’s α = 0.89). We define all reported coefficients, the priors, and the preregistered SESOI on this item-mean scale.

Video game playtime. Digital trace data from Nintendo, Xbox, and Steam provide session-level records (per-title start and end timestamps), while iOS and Android contribute only daily aggregate minutes (no game-level metadata). Nintendo, Xbox, and Steam therefore contribute to both H1 (total playtime) and H2 (genre-specific playtime); iOS and Android enter only the first positive control, which compares self-reported with logged playtime, and the preregistered all-platform H1 model, which Appendix D reports with the other preregistered frequentist analyses (Table A0.1; Appendix A, Deviation D4). Our findings therefore concern play on Nintendo, Xbox, and Steam and do not speak to mobile play.

Game genre classification. We obtained genre labels for Nintendo and Steam games by cross-referencing the game titles in the digital trace data with the Internet Games Database (IGDB). IGDB is, to our knowledge, the only database with complete multi-platform metadata coverage that offers an API for programmatic data retrieval. IGDB crowd-sources its genre labels from a predefined list of options: community members submit contributions (e.g., a new game or an alternative categorisation), which administrators and moderators vet before they appear in the database; IGDB documents its contribution guidelines publicly, including how it holds submissions for review and the predefined list of genres (https://github.com/twitchtv/igdb-contribution-guidelines/wiki). We could not match Xbox titles to IGDB, because Xbox provided us only with anonymised title identifiers; instead, we took the genre labels that Xbox supplies for each title through its store and mapped those onto the IGDB taxonomy. In total, we collected 23 distinct raw IGDB genre labels, converted the Xbox store genres into the IGDB taxonomy, and collapsed them into 14 genre categories (e.g., Strategy subsumes Turn-Based Strategy (TBS), Real-Time Strategy

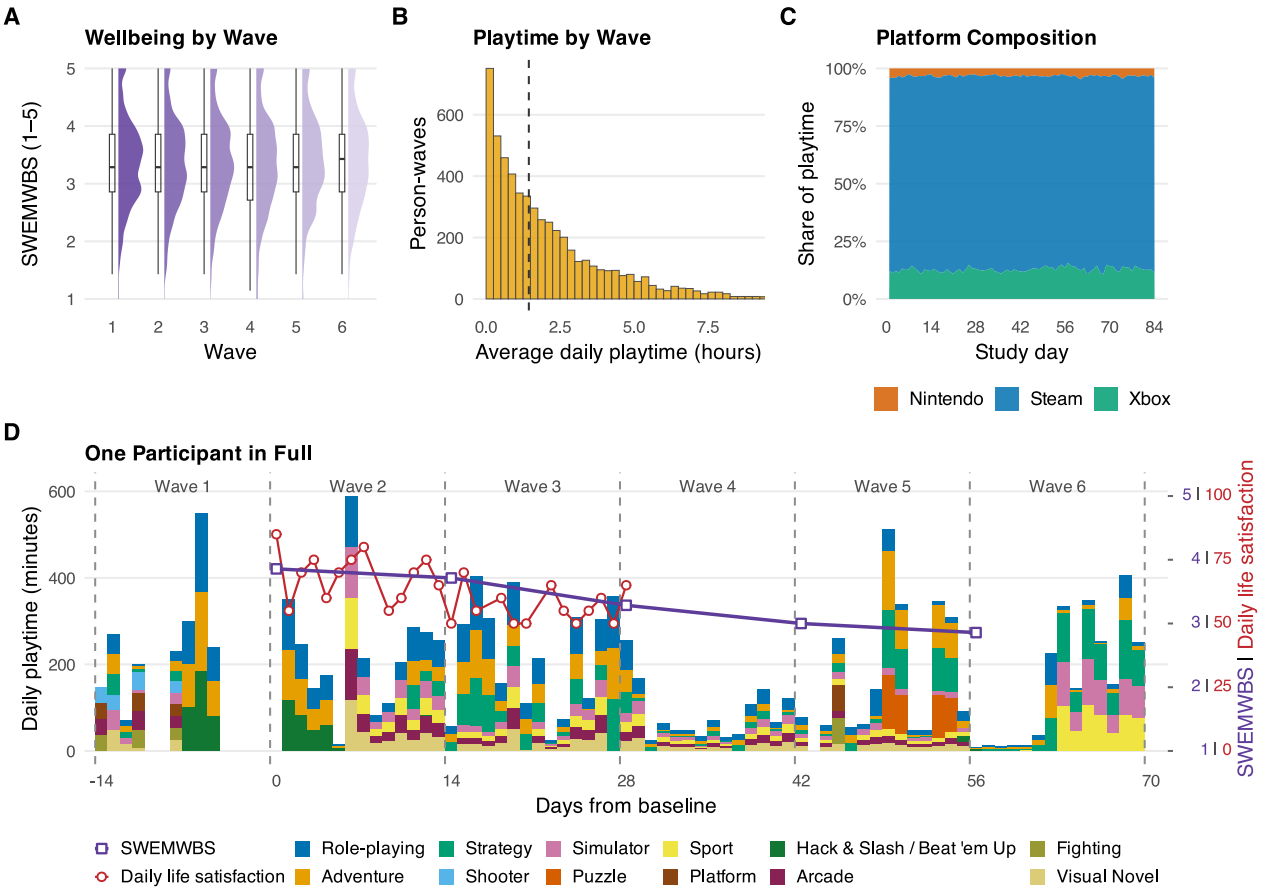


Figure 1. Distributions, Composition, and Raw Structure of the Study Measures. (A) SWEMWBS distributions per biweekly wave (density and median box). (B) Average daily hours per person-wave with recorded play (15-minute bins; dashed line: median). (C) Share of recorded playtime per study day by platform, days 1–84. (D) Daily playtime for one participant (ID p5271179686); bar height is genre-matched playtime, segments are its genre composition under proportional multi-membership weighting. Dashed lines mark the six 14-day waves. Two outcomes are overlaid on the right axis: biweekly SWEMWBS (purple open squares, 1–5) at each survey date, and daily life satisfaction from the 30-day diary (red open circles, 0–100).

(RTS), Tactical, and MOBA; Puzzle subsumes Card & Board Game and Quiz/Trivia). Appendix B gives the complete mapping table, including where the implemented collapsing departs from the preregistered one (Deviation D2), and Appendix C the descriptive statistics by genre. We did not consider IGDB "themes" or other metadata fields. Overall, we matched 93.9% of in-study playtime to at least one study genre (Steam 96.2%, Nintendo 93.1%, Xbox 78.8%); unmatched playtime counts towards total playtime but towards no genre.

Genre co-occurrence and compositional multi-genre attribution. Across the 3,651 individual games in our dataset, 62% carry more than one genre label, leading to 323 distinct genre combinations. Figure 2 shows the playtime-weighted co-occurrence structure across the 40 most-played combinations, which together account for 87% of all genre-matched playtime, and Table 1 lists the ten most-played combinations with their top titles. Because a majority of games carry several labels, the rule that turns a session of a multi-genre game into genre exposure variables decides what a genre coefficient means, so we set out the four rules we considered and the choices we made before describing the models. The preregistration specified the first; the confirmatory model M2 and the exploratory model M3 use the third (Appendix A, Deviation D1); and the exploratory model M4 uses the fourth.

Primary attribution, the preregistered rule, assigns each game to the genre that IGDB lists first and counts a session's full duration towards that genre alone. A 60-minute session of *Skyrim* (Adventure and RPG) would count as 60 minutes of whichever of its two genres IGDB happens to list first and nothing towards the other. Genre exposures then stay mutually exclusive and sum to total playtime, but the rule discards the second and third labels of most games and makes the exposure depend on the order in which IGDB lists them. After data access and before any results were known, we decided not to use it (Deviation D1).

Partial attribution splits a session's playtime across its genres. A 60-minute session of *Skyrim* (Adventure and RPG) would contribute 30 minutes to Adventure and 30 minutes to RPG (or potentially a different, unequal partition). Exposures then still sum to total playtime, but the

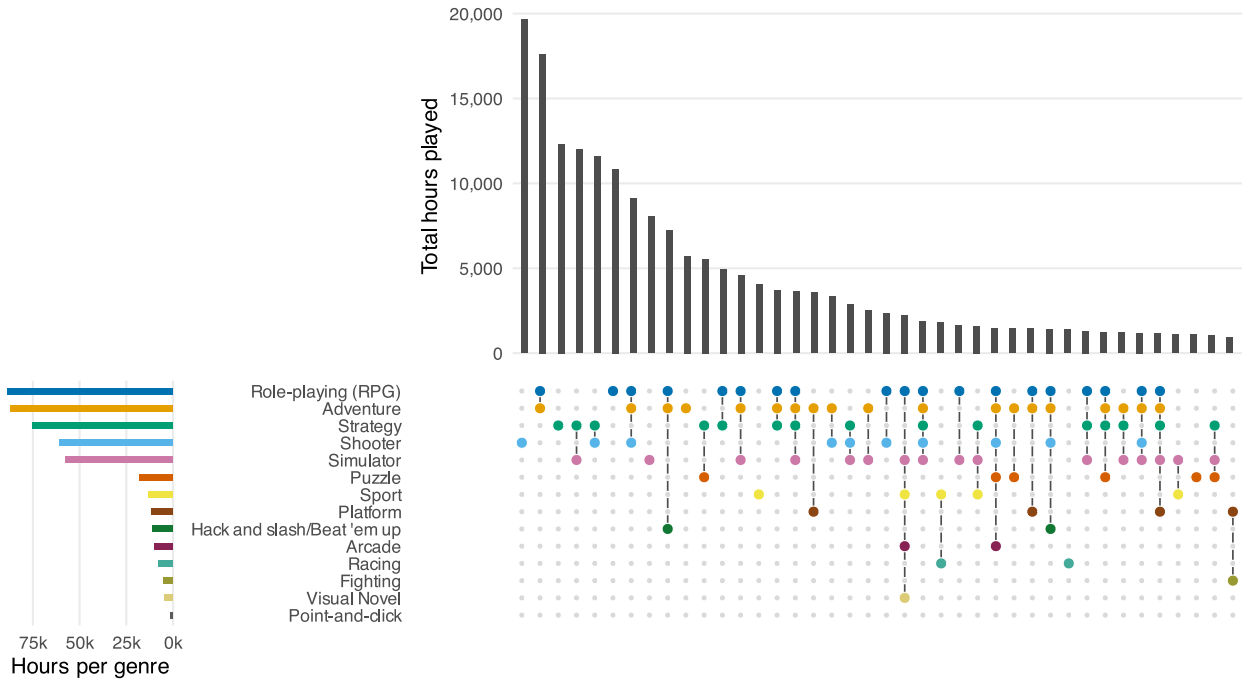


Figure 2. Genre Co-Occurrence Weighted by Total Playtime. Top panel: total hours played by games in each genre combination, ordered by playtime. Bottom panel: dot matrix indicating which genres belong to each combination; connected dots denote multi-genre games. Left bars show total hours attributed to each genre, counting a multi-genre game's full playtime towards each genre exposure.

rule penalises multi-genre games: each additional label a game carries lowers its contribution to every genre, although the game belongs fully to each of its genres rather than partially. We rejected this rule for that reason.

Full attribution, which we used for M2 and M3, counts a session's full duration towards every genre it carries. The same 60-minute *Skyrim* session counts as 60 minutes of Adventure and 60 minutes of RPG. Consider a person who plays 60 minutes of *Skyrim* (Adventure + RPG) and 60 minutes of *Final Fantasy XIV Online* (RPG only). Under partial attribution, *Skyrim* contributes 30 minutes to each genre, yielding 30 total minutes of Adventure and 90 minutes of RPG, a 1:3 ratio implying three times as much RPG engagement as Adventure. Under full attribution, *Skyrim* contributes 60 minutes to each genre, yielding 60 minutes of Adventure and 120 minutes of RPG, a 2:1 ratio that better reflects that RPG-labelled content received twice the engagement of Adventure-labelled content. A consequence is that genre exposures no longer sum to total playtime. Genre-level coefficients therefore represent the association with an added hour of that genre's exposure holding total playtime constant, for example, replacing a single-genre RPG session with a dual-genre Adventure/RPG session of the same length.

Proportional multi-membership weighting, which we used for M4, converts the full-attribution hours into compositional weights by dividing each genre's exposure hours by total exposure (e.g., 33.3% Adventure, 66.7% RPG for a wave consisting of 60 minutes of *Skyrim* and 60 minutes of *Final Fantasy XIV Online*) and uses these as the weights of a multiple-membership structure (Cafri et al., 2015; Fielding & Goldstein, 2006) on a single total-playtime slope. Genre does not enter M4 as an independent predictor of wellbeing but as a modifier of the playtime–wellbeing slope: a wave consisting mostly of RPG play has its playtime coefficient adjusted mostly by the RPG slope adjustment, with other genres contributing in proportion to their share of that wave's play. M4 draws the genre-specific slope adjustments from a common distribution, and the standard deviation of that distribution, σ_s, describes the genre differences that H2 concerns. This weighting shares a limitation with partial attribution, because adding a genre label to a game lowers the shares of the labels it already carries, so none of the four rules is free of trade-offs. In sum, M2 and M3 model full-attribution exposures, M4 models the same hours as proportional weights, and no model in the main text uses the preregistered primary attribution (Appendix D reports the preregistered tests under it), so every genre coefficient below describes exposure to games carrying a label rather than to games filed under it.

Demographics. Participants self-reported age, gender, ethnicity, country, and employment at baseline. Table A0.1 in Appendix C reports full demographic characteristics.

Analytic Approach

We conducted all analyses in R (R Core Team, 2025) and wrote the manuscript in Quarto (Allaire et al., 2025). We fit four Bayesian multilevel models with `brms` (Bürkner, 2017) at the biweekly timescale, and refit each at the

Genre (or combination)	Total hours	% of playtime	n games	Mean min/day	Top 5 titles
Shooter	19661	9	264	13	Marvel Rivals [37.2%], Overwatch 2 [12.2%], Call of Duty: WWII [6.3%], Team Fortress 2 [4.2%], PUBG: Battlegrounds [3.4%]
Adventure + Role-playing	17626	8	210	12	The Elder Scrolls V: Skyrim - Special Edition [7.6%], Clair Obscur: Expedition 33 [6.9%], The Elder Scrolls IV: Oblivion Remastered [5.6%], Elden Ring [5.6%], Throne and Liberty [4.7%]
Strategy	12322	6	136	8	Dead by Daylight [29%], R.E.P.O. [17.1%], Dota 2 [10.7%], DeadLock [7%], Bloons TD 6 [6.9%]
Simulator + Strategy	11984	6	202	8	RimWorld [19.3%], Schedule I [16.5%], Europa Universalis IV [6.7%], Hearts of Iron IV [6.3%], Victoria 3 [4.2%]
Shooter + Strategy	11592	6	34	8	Counter-Strike 2 [42.6%], Helldivers 2 [16.7%], Apex Legends [13%], 5268BC7F… [9.8%], Tom Clancy's Rainbow Six Siege X [6.9%]
Role-playing	10840	5	133	7	Elden Ring: Nightreign [24.1%], Final Fantasy XIV Online [22.7%], Lightning Returns: Final Fantasy XIII [6.7%], Star Wars: The Old Republic [5.1%], 6007952B… [3.3%]
Adventure + Role-playing + Shooter	9113	4	49	6	Warframe [21.3%], Destiny 2 [16.1%], Cyberpunk 2077 [10.3%], Rust [8.8%], Red Dead Redemption 2 [8.5%]
Simulator	8070	4	211	5	7DAC9E96… [15%], Animal Crossing: New Horizons [11.3%], Garry's Mod [9.1%], CAC402B8… [6.6%], House Flipper 2 [5%]
Adventure + Hack and slash/Beat 'em up + Role-playing	7224	4	51	5	Path of Exile [30.9%], Monster Hunter Wilds [19.3%], Path of Exile 2 [12%], Torchlight: Infinite [5.2%], Lost Ark [4.3%]
Adventure	5728	3	346	4	2089FA57… [8.5%], C3A5EDE3… [6.2%], F4FF1A75… [5%], C1E17D99… [4.8%], 79207D00… [3.9%]

Table 1. Top 10 genre combinations by total playtime. Multi-genre rows join labels with ' + '. Total hours = playtime for games with that combination; % of playtime = share of all genre-matched playtime; n games = unique game-platform pairs. Xbox titles are anonymised (and truncated for this table).

daily timescale as an exploratory extension. Sampling ran in CmdStan (version 2.39.0) through `cmdstanr`, with four chains per model and within-chain parallelisation, which affects only computation time. We preregistered two of them: the model of total playtime (M1) tests H1, and the genre model with independently estimated genre coefficients (M2) tests H2. We added the other two after seeing the M2 results, to examine their robustness: M3 keeps M2′s predictors but pools the genre coefficients through a common prior, whereas M4 keeps total playtime as the single predictor, as in M1, and lets each wave's genre composition modify its slope; both describe the size of the genre differences through the estimated spread of the genre effects, and M4 also yields H1 estimates. All four models use the `brms` default priors, and Appendix F gives their equations and priors in full. We organise the results by hypothesis and describe the preregistered models first and the exploratory models second, and we label every figure row, legend entry, and table row by model, with its preregistration status. We catalogue deviations from the preregistered plan in Appendix A and flag them below where they affect interpretation, and we report the preregistered frequentist analyses in full in Appendix D.

Smallest Effect Size of Interest

Establishing the absence of a practically significant effect requires committing in advance to a threshold below which an association is too small to matter, the smallest effect size of interest (SESOI). By practically significant, we mean the smallest change in wellbeing that a player would themselves notice, known also as a minimally important difference (MID). Following previous studies, we set the MID at 0.3 points on the SWEMWBS item-mean scale (1–5; equivalent to 2.1 points on the 7–35 summed scale), converging with anchor- and distribution-based estimates of the Warwick-Edinburgh Mental Wellbeing Scale (WEMWBS) (Ballou, Sewall, et al., 2024; Maheswaran et al., 2012) and the Positive and Negative Affect Schedule (PANAS) (Anvari & Lakens, 2021).

To calculate the SESOI, we also need to know how large a change in playtime is realistically available to reach the MID. Because US adults average approximately 5 hours of daily leisure time (Sturm & Cohen, 2019), the figure to which we anchored the preregistered SESOI, we take 5 hours as an upper bound on how much daily play could plausibly rise or fall without displacing other behaviours that arguably impact wellbeing more than play (e.g., sleep). More recent diaries put the figure lower for our age range. Americans aged 23 to 38 averaged about 4.3 hours a day in 2019 (Freeman, 2022), and 3.8 to 4.7 hours

by age band between 20 and 44 in 2024 (U.S. Bureau of Labor Statistics, 2025); UK adults aged 18 to 44 averaged between 4.3 and 5.2 hours in 2014, depending on sex and age band (summing the leisure components in Table 3 of García Román & Gracia (2022)). Five hours is therefore an upper bound, and the preregistered SESOI the stricter of the two choices. Our SESOI is the ratio of the two: 0.3 ÷ 5 = 0.06 points per hour, the same threshold used by Ballou, Sewall, et al. (2024). An association smaller than 0.06 points per hour means that even a person who shifted their entire 5-hour daily leisure budget into or out of gaming would not experience a noticeable change in wellbeing.

We evaluate the individual coefficients, the two total-playtime coefficients of H1 and each genre's within-person and between-person coefficients, against the interval between the negative and positive SESOI [−0.06, +0.06], also known as the Region of Practical Equivalence (ROPE) (Kruschke, 2018). If a coefficient's 95% credible interval (CI) falls entirely inside the ROPE, we consider the effect practically null, meaning even the largest plausible effect is too small to matter; a CI entirely beyond either bound indicates a practically positive or negative effect, and one straddling either edge is inconclusive. Throughout, CI denotes a 95% credible interval from a Bayesian model; the preregistered frequentist tests in Appendix D use 90% confidence intervals, as the equivalence test requires, and are labelled as such, so the interval level identifies the type. This rule replaces the preregistered decision grid for H1 (Deviation D3), which classifies the 90% confidence interval of each H1 coefficient as follows: an interval entirely beyond either bound indicates a practically significant positive or negative association; one that excludes zero and crosses a bound, a positive or negative association that may be practically significant; one that excludes zero and lies inside the bounds, a positive or negative association that is not practically significant; one that covers zero and lies inside the bounds, no practically significant association; and one that covers zero and crosses a bound, an inconclusive result. Appendix D reports the preregistered analyses under this grid.

We test H2 with the preregistered joint Wald tests of whether the genre coefficients are equal, which we run separately for the 14 within-person (H2a) and the 14 between-person (H2b) coefficients on a maximum-likelihood fit of M2 with the same fixed effects and random intercept. These tests answer H2, and we report them at the start of the H2 results. A rejection shows that at least two genres differ but not by how much, and the preregistration fixed no threshold for the size of differences between genres. We therefore describe the size of the differences without converting it into a verdict. For each model we compute the spread *s* of the genre effects at each posterior draw: as the standard deviation of M2′s 14 independently estimated coefficients (which includes their estimation error and can therefore overstate the spread), as τ_{genre} in M3, and as σ_s in M4. Treating the genre effects as normally distributed around their average, we translate *s* into three quantities and summarise each in the text by its posterior mean and 95% CI; tables and the daily section report posterior means. The first is the share of genres whose effect lies more than 0.06 from the average genre effect, $2[1 - \Phi(0.06/s)]$, where Φ is the standard normal distribution function, together with the expected number of our 14 genres that this share implies. The second is the share of genre pairs whose effects differ by more than 0.06, $2[1 - \Phi(0.06/(\sqrt{2}\,s))]$, because the difference between two genre effects drawn from the same distribution has standard deviation $\sqrt{2}\,s$. The third is the share of the genre-effect distribution, a normal distribution with the model's average genre effect as its mean and *s* as its standard deviation, that lies outside the ROPE (Vuorre, Kay, et al., 2024). Unlike the first two, this share depends on the average genre effect as well as on the spread, so it describes whether the associations are practically null in every genre rather than whether genres differ. We do not turn these quantities into a verdict, because we can translate the SESOI into a threshold for differences between genres in more than one defensible way, and the translations set different thresholds: for 95% of genre effects to lie within 0.06 of the average, the spread must be below 0.031 (0.06 ÷ 1.96), whereas for 95% of genre pairs to differ by less than 0.06, it must be below 0.022 (0.06 ÷ (1.96 × √2)). We report both values as reference points only.

Preregistered Models

M1: Total Playtime

M1 is the preregistered model for H1: is total playtime across Nintendo, Xbox, and Steam associated with wellbeing? We split each participant's playtime into a between-person part, their mean across waves, and a within-person part, each wave's deviation from that mean, and regress SWEMWBS on both, so the model returns two coefficients: β_W, the association between a wave's deviation from a person's usual playtime and that wave's wellbeing, and β_B, the association between a person's habitual playtime and their average wellbeing. We estimate a random intercept for each participant because people differ in baseline wellbeing, and a random slope for within-person playtime because the association need not be identical for everyone (equations and priors in Appendix F).

M2: Independent Genre Coefficients

M2 is the preregistered specification for H2. We estimate a separate within-person and between-person coefficient for each of the 14 genres, independently of one another, with a random intercept per participant. Because genre exposures are non-exclusive, their sum does not recover total playtime, and M2 therefore contains no separate total-playtime term. We departed from the preregistered specification in two respects. First, a game's full playtime counts towards every one of its genre labels (Deviation D1), whereas the preregistration assigned each game its first IGDB genre only; this adds no coefficients for genre combinations, since there is still one coefficient per genre and dimension, but the 14 exposure columns overlap. Second, the preregistration specified per-participant random slopes for the within-person genre terms, with all random-effect correlations fixed at zero. When we fitted this structure by maximum likelihood to M2′s data, it was singular, with most of the 14 slope standard deviations estimated at

zero, as the preregistration anticipated, so we retained the random intercept only (equations and priors in Appendix F).

Exploratory Models

We did not preregister the models in this section and report them as exploratory throughout, together with the daily-timescale refits described at the end of this section.

M3: Hierarchical Prior

For M3 we keep M2′s predictors and random intercept but use a multilevel model that treats the 14 genre-specific coefficients as draws from a common normal distribution rather than as unrelated quantities, and we estimate both the centre and the standard deviation of that distribution, separately for the within-person and between-person dimensions. The centres, γ_W and γ_B, are the common genre effects, the average association of genre playtime with wellbeing around which the genre-specific coefficients vary. The standard deviations, τ_{genre} (τ_{within} and $\tau_{between}$), measure how widely the genre coefficients spread around these common effects, which is the heterogeneity H2 targets. In the Results we report the common genre effects beside the mean of the 14 independently estimated M2 coefficients, their unpooled counterpart. The shared distribution also produces partial pooling: it pulls the estimates for rarely played genres towards the common genre effect, whereas genres with ample data retain estimates close to their independent values. Per-genre intervals from M3 therefore partly reflect the data on the other genres. An earlier version of M3 fixed the centres of these distributions at zero; following co-author review, and after the results of that version were known, we estimate them instead, so that τ measures the spread around the common genre effect rather than around zero. Because sampling near $\tau = 0$ is numerically difficult, we estimate M3 in a non-centred form that is mathematically equivalent to sampling each coefficient directly, use conservative sampler settings, and report divergence counts and R-hat diagnostics for every model alongside the results (equations and priors in Appendix F).

M4: Multi-Membership

M4 describes each wave by its total playtime and by the genre composition of that playtime, the representation that panel D of Figure 1 uses for days. The model keeps total playtime as the predictor and inherits M1′s structure, including the overall β_W and β_B, and lets the composition of that playtime modify the playtime slope. Each genre adjusts a wave's slope in proportion to its share of the wave's genre exposure, so a wave consisting mostly of Shooter play is governed mostly by the Shooter slope adjustment, and a wave divided between Adventure and RPG draws on both adjustments in proportion. As described under the attribution rules, M4 draws the genre adjustments from a common distribution whose standard deviation, σ_s (within and between persons), quantifies how much the playtime–wellbeing slope varies with the genres being played (equations, priors, and a parameter-recovery check in Appendix F).

Daily Timescale

To assess whether our findings hold on a shorter timescale, we conducted an exploratory daily-timescale analysis, whose plan we fixed after the preregistered analyses were complete but before we estimated any daily model. We made two changes after the daily results were known: we estimated M3-D's centres as in M3, and we gave all daily models the `brms` default priors (Appendix F) instead of the plan's priors, which had rescaled the biweekly priors to the life-satisfaction slider. The Open Play study asked US participants to complete daily surveys for the first 30 study days, including a life-satisfaction item ("I was satisfied with my life today") answered on a 0–100 slider (Ballou, Földes, Vuorre, et al., 2025), which serves as the outcome. The exposure is console and PC playtime from 4:00 a.m. local time of the response's waking day up to the survey completion timestamp, which we computed from the same session-level telemetry. Of the 14,814 diary responses from 956 participants in the analytic sample, we analysed the 13,338 responses (90%), from 749 participants, for which telemetry and an analysable same-day window were available. We again split playtime into a within-person and a between-person part, but measure the two parts over different windows. The within-person part is the difference between the day's same-day playtime and the participant's own average same-day playtime. The between-person part is the participant's habitual playtime over a full day, so that differences between participants remain expressed in the same daily-hour units as in the biweekly analysis. We construct the SESOI for the slider in the same way as the preregistered one, as a minimally important difference divided by the roughly five hours of daily leisure realistically available (Sturm & Cohen, 2019). No anchor-based minimally important difference exists for a single-item life-satisfaction slider, so we set it at 5 slider points, a deliberately conservative choice. It lies below two available benchmarks, half the within-person SD of the slider (6.1 points) and the preregistered minimally important difference expressed as a share of its scale range (7.5% of 0–100, or 7.5 points), and a smaller minimally important difference makes practical equivalence harder to claim rather than easier. Dividing by five hours gives 1 slider point per hour of daily play. We refit the model specifications (M1–M4) at the daily level and label them M1-D to M4-D. They use the same `brms` default priors as the biweekly models, so `brms` sets the intercept location and the standard-deviation scales from the life-satisfaction outcome: the intercept prior is centred at its median, and the scale of the intercept prior and of every standard-deviation prior, including τ in M3-D, is the larger of 2.5 and its median absolute deviation. Appendix D details window construction, sample exclusions, and four robustness refits.

Missing Data

All missingness in the outcome is wave nonresponse, because every completed survey contains all seven SWEMWBS items. Survey completion declined from 100% at wave 1 to 47% at wave 6, leaving 3,490 of the 10,008 person-waves (34.9%) without an observed

SWEMWBS. The dataset paper documents retention over the study period and missingness by data source (Ballou, Földes, Vuorre, et al., 2025).

We fit the models to the observed outcomes, an approach that is valid provided outcome missingness is at random given the modelled covariates and random effects. To test that assumption, we multiply imputed SWEMWBS on the complete participant × wave grid with two-level multiple imputation by chained equations (`mice` and `miceadds`) (Robitzsch & Grund, 2024; van Buuren & Groothuis-Oudshoorn, 2011): predictive mean matching (`2l.pmm`) with participants as random-intercept clusters (van Buuren, 2018), and m = 100 imputations, a number we set from the pilot fraction of missing information by von Hippel's (2020) two-stage rule. Predictive mean matching donates observed values, so imputed scores respect the 1–5 scale. We fitted frequentist counterparts of the analysis models to each imputed dataset and pooled the estimates by Rubin's rules; neither pooled H1 coefficient was practically significant, as in the complete-case analyses, whereas the pooled tests no longer reject equality of the within-person genre coefficients (Appendix D, Table A0.4); Table A0.1 reports the preregistered complete-case tests. We did not impute playtime because tracking was continuous and automatic even for participants with missing survey waves, so within linked platforms an absence of play is recorded absence rather than missing data, and exposures exist for every person-wave.

Results

The analytic sample comprised 1,668 participants with a valid SWEMWBS response in at least one biweekly wave, yielding 6,518 person-wave observations (M = 3.9 of 6 waves per participant; see Missing Data). We measured playtime from linked Nintendo, Xbox, and Steam telemetry; participants played an average of 1.75 hours per day (SD = 2) across the three platforms.

We report the preregistered data quality controls first and then each hypothesis in turn. Within each hypothesis section, the preregistered model (M1 for H1, M2 for H2) comes first and the exploratory models (M4 for H1; M3 and M4 for H2) second, and figure rows, legend entries, and table rows are labelled by model and preregistration status. The exploratory daily-timescale refits and the robustness checks close the section.

Data Quality Controls

Before testing the hypotheses, we ran the three preregistered positive controls, each of which checks that the data reproduce a pattern known to hold.

First, self-reported biweekly playtime correlated positively with digital trace playtime (r = 0.481, 95% confidence interval [0.459, 0.502], $p < .001$).

Second, session integrity largely held. Xbox contained 0 overlapping sessions, and no Steam session's recorded playtime exceeded its elapsed wall-clock duration (0 violations). Nintendo retained a small number of overlapping sessions: 717 of 322,662 sessions (0.22%), with a median overlap of 9.8 minutes and accounting for 0.09% of total Nintendo playtime.

All overlapping sessions could be traced to concurrent use of multiple consoles linked to one account (e.g., a Switch Lite overlapping an OLED model). Given their rarity, we judged them neither a sign of systematic data quality problems nor frequent enough to affect the estimates.

Third, men played more shooter games than women (M = 0.511 vs. 0.327 hrs/day, $t(1011.5) = 4.2$, $p < .001$, Cohen's $d = 0.22$), meeting the preregistered criterion of a significant difference in the expected direction and consistent with population studies of gendered genre preferences (Rehbein et al., 2016). All three controls passed, and we proceeded to the hypothesis tests.

Total Playtime (H1)

H1 predicted that total playtime is not meaningfully associated with wellbeing, whether a person plays more or less than usual in a given wave (H1a) or habitually plays more or less than other people (H1b). The preregistered model is M1, the within-between regression of total playtime; we judged each of its two coefficients against the ROPE of ±0.06 points per hour/day. M1 was estimated with Bayesian methods rather than the preregistered frequentist equivalence test (Deviation D3). Under the preregistered decision grid, the preregistered frequentist test reported in Appendix D likewise classified neither coefficient as practically significant. Figure 3 shows the posterior distributions, and Appendix E (Table A0.1) tabulates all hypothesis tests.

Preregistered Model (M1)

Within persons (H1a), the posterior mean was 0.004 SWEMWBS item-mean points per hour/day (95% CI: [−0.007, 0.015]), with >99.9% of the posterior draws inside the ROPE. In waves in which a participant played more than their own average, wellbeing did not differ by a practically meaningful amount.

Between persons (H1b), the posterior mean was −0.015 (95% CI: [−0.034, 0.003]), with >99.9% of the posterior draws inside the ROPE. The estimate leaned negative, with a credible interval that reached zero, but its magnitude was far below the SESOI: a 2 hour/day difference in habitual playtime corresponded to −0.031 points on the 1–5 scale, about 0.8% of the 4-point range.

Both components of H1 were supported under the ROPE rule (Deviation D3) and under the preregistered 90% CI grid (Appendix D), which classified each coefficient as showing no practically significant association.

M1′s random slope estimates how much the within-person association varies between participants. The SD of the person-specific associations was 0.029 (95% CI: [0.003, 0.058]), and 1.8% of its posterior lay above the SESOI. Under the model's normal random-slope assumption, this spread combined with the average association implied that the within-person association exceeded the SESOI, in either direction, for about 8.1% of participants. The Discussion returns to this minority.

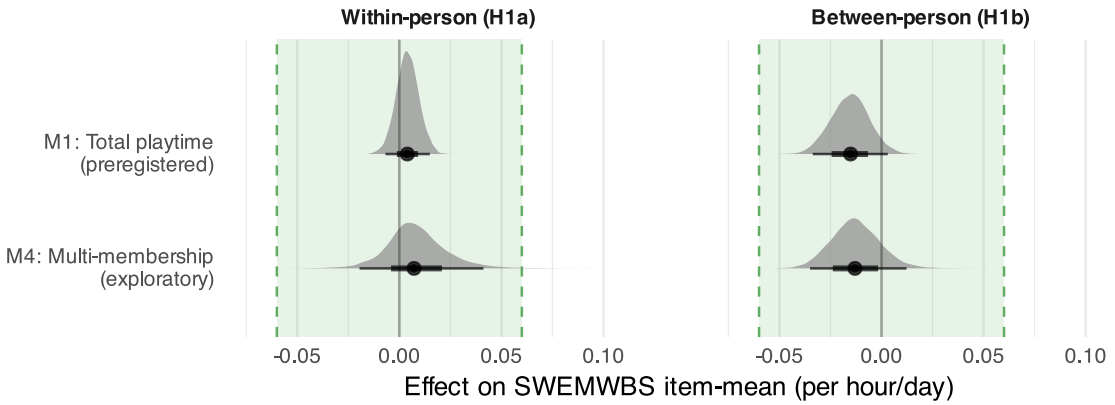


Figure 3. Posterior Distributions for Total Playtime Effects (H1). M1 (top row) is the preregistered total-playtime model; M4 (bottom row) is the exploratory multi-membership model, which keeps total playtime as its predictor and lets genre composition modify the slope. M2 and M3 have no separate total-playtime predictor and are excluded. Green band: ROPE (±0.06). Thick: 66% credible interval; thin: 95% CI.

Exploratory Model (M4)

M4 belongs to H2, but its fixed effects are H1 quantities: the average within-person and between-person associations of total playtime, estimated without assuming that the association is the same for every genre. We report them here beside M1′s; the genre deviations themselves describe the genre differences that H2 concerns and are reported in the next section. M4′s estimates matched M1′s. Within persons, β_W = 0.008 (95% CI: [−0.019, 0.041]), with 99.6% of the posterior draws inside the ROPE; between persons, β_B = −0.013 (95% CI: [−0.035, 0.012]), with >99.9% of the draws inside (Figure 3). Allowing the association to depend on which genres were played left the total-playtime association unchanged. M2 and M3 have no total-playtime term and do not bear on H1.

Genre Differences (H2)

Our second hypothesis was that genres differ in how playtime relates to wellbeing within persons (H2a) and between persons (H2b). The preregistered model is M2, which estimates a within-person and a between-person coefficient for each of the 14 genres. The exploratory models M3 and M4 estimate the spread of the genre effects as a model parameter. Figure 4 and Figure 5 show these results across the three models.

Preregistered Model (M2)

The preregistered inference for H2 is a joint Wald test of whether the 14 genre coefficients are equal. We fitted a frequentist version of M2 by maximum likelihood, with the same fixed effects and random intercept, and tested the equality of its coefficients. The null hypothesis sets the 13 differences between the first genre's coefficient and each of the others to zero, which holds only if all 14 coefficients are equal; rejecting it means that at least one difference is non-zero, and hence that at least two genres differ. Within persons, the test rejected equality ($\chi^2(13)$ = 24.8, p = 0.025): at least two genres differed in how a change in their playtime related to a change in wellbeing. Between persons, the test did not reject equality ($\chi^2(13)$ = 14.9, p = 0.315). Under the preregistered criteria, H2a was supported and H2b was not. Appendix D reports these tests together with the preregistered H1 equivalence tests, also under two preregistered specifications (Table A0.1); under the preregistered primary-genre attribution, the within-person test did not reject equality (p = 0.216).

The joint Wald test shows whether the coefficients differ, but not which ones differ or by how much. To describe the size of the differences from the same model, we computed the SD of the 14 genre coefficients at each posterior draw of the Bayesian fit of M2, and from it the descriptors defined in the Methods; none of them was preregistered, and none was converted into a verdict. Within persons, the SD averaged 0.032 (95% CI: [0.02, 0.047]; Figure 4, top row). It implied that 7% of genres (95% CI [0.3%, 20.3%]) lay more than 0.06 from the average genre effect, an expected 0.98 of the 14, and that 18.5% of genre pairs (95% CI [3.7%, 36.8%]) differed by more than 0.06. Between persons, the SD averaged 0.082 (95% CI: [0.048, 0.129]), which implied that 44.3% of genres (95% CI [21%, 64.1%]) lay more than 0.06 from the average genre effect, an expected 6.2 of the 14, and that 58.4% of genre pairs (95% CI [37.5%, 74.1%]) differed by more than 0.06. Because M2 estimates the 14 coefficients independently, each carries its own estimation noise, which is large for rarely played genres, and the SD of the 14 estimates, and the shares derived from it, can therefore overstate how much the genres truly differ.

Figure 5 (blue circles) shows the per-genre estimates from M2. The 14 within-person point estimates clustered near zero, and 8 of the 14 credible intervals fell entirely inside the ROPE; the remainder straddled a boundary rather than excluding it. The cost of independent estimation showed in the interval widths: the widest within-person interval (Point-and-Click) was about 5 times wider than the narrowest, too wide to support a conclusion about any rarely played genre. Between persons the intervals were wider still, with 2 of the 14 entirely inside the ROPE.

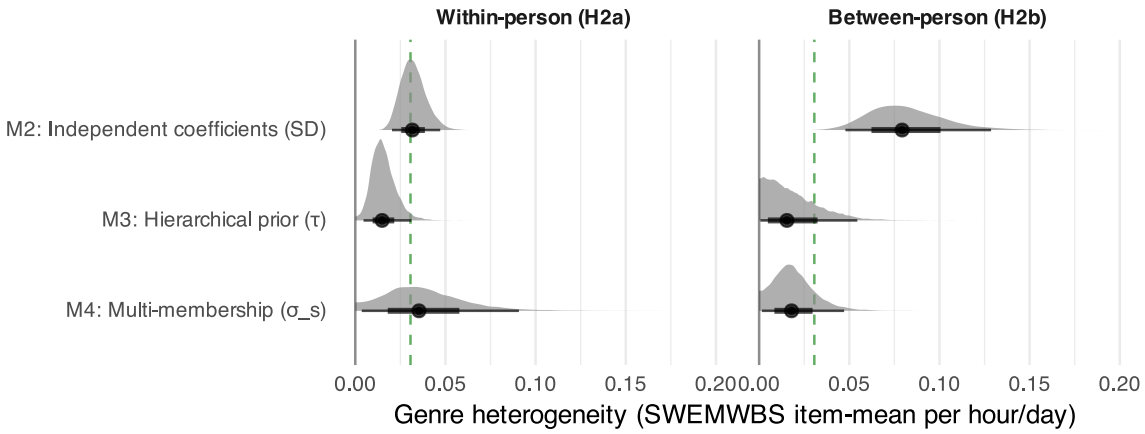


Figure 4. Genre Heterogeneity Under Three Model Specifications (H2). Left: within-person (H2a); right: between-person (H2b). M2 (top row) is the preregistered model (SD of its 14 independently estimated coefficients, inflated by their estimation noise); M3 (middle row) and M4 (bottom row) are exploratory (τ and σ_s, model-estimated spreads). Dashed line: reference value of 0.031, the spread at which 95% of genre effects would lie within 0.06 of the average genre effect; the corresponding value for 95% of genre pairs differing by less than 0.06 is 0.022. Both are reference points, not decision criteria (Methods). Thick: 66% credible interval; thin: 95% CI.

Exploratory Models (M3 and M4)

M3 and M4 were specified after the M2 results were known. Under M3, the hierarchical prior model, the spread of the within-person genre effects (H2a) was τ_{within} = 0.016 (95% CI: [0.005, 0.031]; Figure 4, middle row). This spread implied that 0.6% of genres (95% CI [0%, 5.4%]) lay more than 0.06 from the common genre effect, an expected 0.08 of the 14, and that 2.6% of genre pairs (95% CI [0%, 17.3%]) differed by more than 0.06. Between persons (H2b), $\tau_{between}$ = 0.019 (95% CI: [0.001, 0.054]), which implied that 3.4% of genres (95% CI [0%, 27%]) lay more than 0.06 from the common genre effect, an expected 0.48 of the 14, and that 7.9% of genre pairs (95% CI [0%, 43.6%]) differed by more than 0.06. The prior on τ places 0.9% of its mass below the reference value of 0.031, against 97.2% of the posterior for τ_{within}, so the concentration of τ_{within} at small values came from the data rather than the prior. M3 estimated the common genre effects at γ_W = 0.003 within persons (95% CI: [−0.008, 0.015]) and γ_B = −0.008 between persons (95% CI: [−0.026, 0.011]); the means of the 14 independently estimated M2 coefficients were 0.007 (95% CI: [−0.003, 0.018]) and −0.013 (95% CI: [−0.039, 0.014]), respectively.

Under M4, the multi-membership model, the spread of the within-person genre adjustments (H2a) was $\sigma_{s,\ \text{within}}$ = 0.038 (95% CI: [0.004, 0.091]; Figure 4, bottom row). This spread implied that 14.1% of genres (95% CI [0%, 50.9%]) deviated from the overall playtime slope by more than 0.06, an expected 2.0 of the 14, and that 24.8% of genre pairs (95% CI [0%, 64%]) differed by more than 0.06; spreads up to the upper credible bound of 0.091 points per hour/day remained plausible. Between persons (H2b), $\sigma_{s,\ \text{between}}$ = 0.019 (95% CI: [0.002, 0.047]), which implied that 2.5% of genres (95% CI [0%, 20.3%]) deviated from the overall playtime slope by more than 0.06, an expected 0.35 of the 14, and that 6.9% of genre pairs (95% CI [0%, 36.8%]) differed by more than 0.06.

The three estimates in Figure 4 measure the spread of the genre effects in different ways. M2 estimates each genre independently, so its 14 coefficients are noisy and the SD across them can overstate the spread. M3 and M4 instead pool information across genres: a hierarchical model shrinks the noisier estimates towards the common centre, which reduces the impact of noise and of multiple comparisons (Gelman et al., 2012). Within persons, M4′s σ_s was less precise than M3′s τ, because it describes how the playtime slope changes with the genre composition of a wave rather than how separate genre exposures differ.

M3 shrinks the per-genre estimates towards the common genre effect (Figure 5, orange squares): 14 of the 14 within-person credible intervals fell entirely inside the ROPE, as did 10 of the 14 wider between-person intervals.

M4 estimates each genre as a deviation from the overall playtime slope (Figure 5, purple diamonds). With the overall fixed effect added back, its within-person estimates spanned −0.021 to 0.038 and its between-person estimates −0.034 to −0.003.

The spread describes how far the genre effects lie from one another. Whether the associations are practically null in every genre depends on where the genre effects are centred as well as on their spread, so we also computed, at each posterior draw, the share of each model's genre-effect distribution lying outside the ROPE (Vuorre, Kay, et al., 2024). Within persons, that share was 7.9% under M2 (95% CI: [0.4%, 22.5%]), 0.7% under M3 (95% CI: [0%, 7%]), and 16.3% under M4 (95% CI: [0%, 54%]); between persons, it was 45.4% under M2 (95% CI: [22.2%, 65%]), 4.2% under M3 (95% CI: [0%, 30.7%]), and 4.1% under M4 (95% CI: [0%, 26.1%]). Within persons, the share was smallest under M3; the M2 shares were inflated by the wide intervals of rarely played genres, and the M4 shares reflected uncertainty in both the spread and the overall playtime slope.

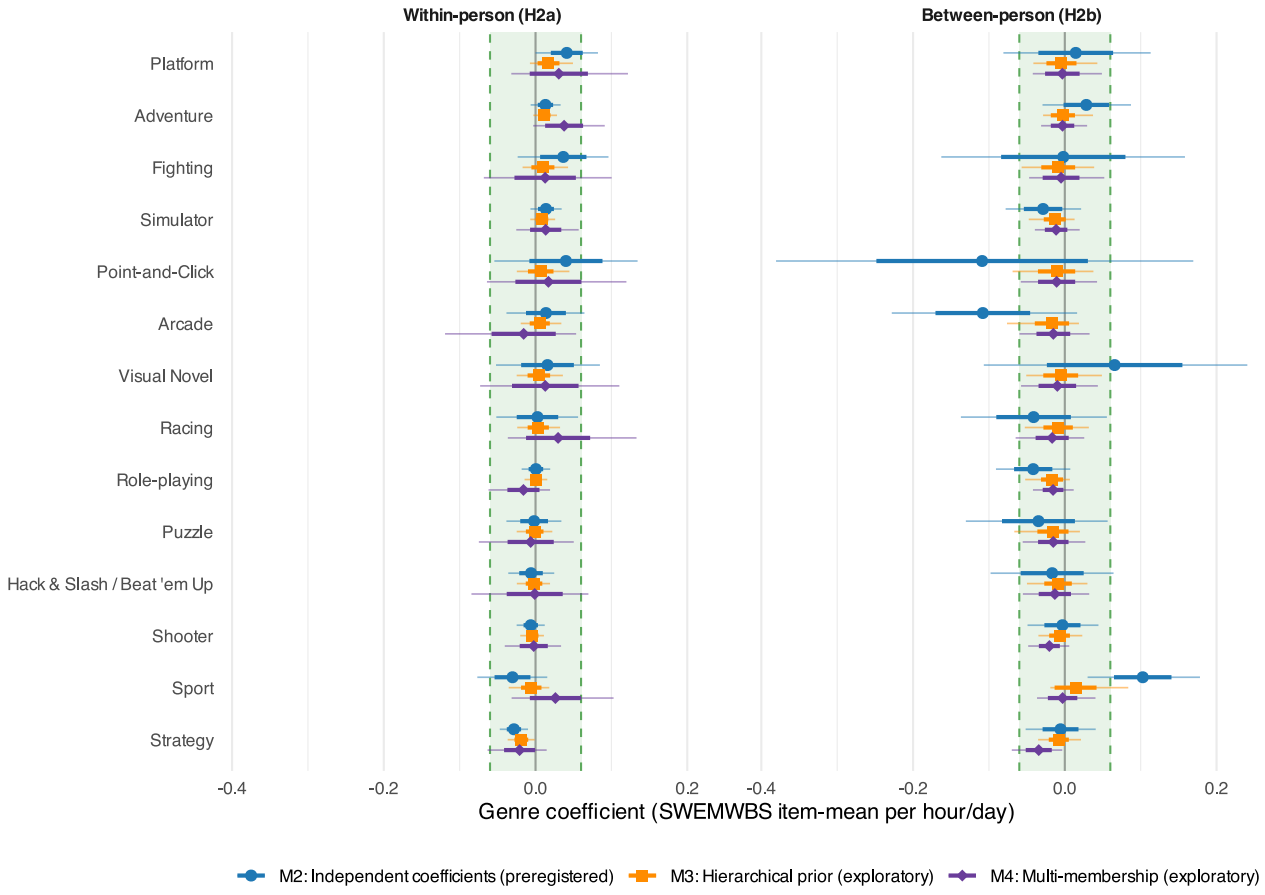


Figure 5. Per-Genre Exposure Coefficients Under Three Model Specifications (H2). Left: within-person (H2a); right: between-person (H2b). Each genre row shows M2 (blue circle; preregistered model, no pooling), M3 (orange square; exploratory, hierarchical prior), and M4 (purple diamond; exploratory, multi-membership). The M2 series is the caterpillar plot specified in the preregistration. Genres ordered by the M3 within-person estimate. Green band: ROPE (±0.06). Thick bars: ±1 posterior SD; thin bars: 95% CI.

In sum, the preregistered tests answered H2: the within-person genre coefficients were not all equal, so H2a was supported (although not under the preregistered primary-genre attribution; Appendix D), and the between-person test did not detect differences, so H2b was not. The spreads described how large the within-person differences were. Under M3, an expected 0.08 of the 14 genres lay more than 0.06 from the common genre effect and 2.6% of genre pairs differed by more than 0.06; the corresponding figures were 0.98 genres and 18.5% of pairs under M2, whose SD includes estimation noise, and 2.0 genres and 24.8% of pairs under M4, whose spread was less precisely estimated. As set out in the Methods, we do not convert these quantities into a verdict on the practical size of the genre differences. Between persons, M2′s SD was inflated by the noise in the wide between-person intervals: its upper credible bound was more than twice that of M3′s $\tau_{between}$. Genre terms did not improve out-of-sample prediction (Appendix D), and the wide intervals for rarely played genres ruled out confident inference about any single genre. Appendix E (Table A0.1) tabulates the joint tests, the spread estimates and their descriptors, and the convergence diagnostics for the three models; full posterior summaries for all four models are in the supplementary materials.

Daily Timescale (Exploratory)

The daily-level analyses (M1-D to M4-D) were not preregistered, and every estimate in this section is exploratory. Participants in the analysed daily sample contributed a median of 21 diary days, with participation declining from 956 responses on day 1 to 398 on day 30; windows closed at the survey completion timestamp (median length 13.7 hours), so 60.7% of person-days recorded no console or PC play.

Figure 6 displays the posteriors against the ROPE. At the daily level, the within-person association between same-day playtime and life satisfaction was positive, at β = 0.17 points per additional hour in the standalone total-playtime model (M1-D; 95% CI [0, 0.34]), with the lower credible bound at zero and >99.9% of the posterior draws inside the ROPE; the multi-membership model's fixed effect was similar (M4-D: β = 0.13, 95% CI [−0.31, 0.51]). The interval widened past zero under two of the three robustness refits (Appendix D). Person-to-person variation in this slope was practically negligible: the random-slope

SD was 0.7 (95% CI [0.28, 1.01]), with 97.2% of the posterior below the SESOI. Between persons, with habitual play measured in full-day hours, participants who played more than others reported slightly lower daily life satisfaction: M1-D gave $\beta = -0.4$ points per habitual daily hour (95% CI [−1.15, 0.29]), a negative lean whose posterior sat 94.7% inside the ROPE; M4-D gave $\beta = -0.4$ (95% CI [−1.11, 0.3]); and the all-responses refit, which scores unobserved play as zero, gave $\beta = -0.36$ (95% CI [−1.02, 0.28]) (Table A0.2).

At the daily level, we described the genre spreads with the same quantities as at the biweekly level, with the daily SESOI of one slider point per hour in place of 0.06, and again without a verdict; the corresponding reference values are 0.51 slider points ($1 \div 1.96$) for genre effects around their average and 0.36 ($1 \div (1.96 \times \sqrt{2})$) for genre pairs. Within persons, the hierarchical prior gave $\tau_{within} = 0.22$ (95% CI [0.01, 0.57]), which implied an expected 0.10 of the 14 genres more than one point from the common genre effect and 2.7% of genre pairs more than one point apart. The multi-membership model gave $\sigma_{s,within} = 0.49$ (95% CI [0.03, 1.3]; an expected 1.2 genres and 16.6% of pairs), and the per-draw SD of M2-D's 14 independently estimated coefficients, inflated by estimation error as at the biweekly level, was 0.63 (95% CI [0.41, 0.87]; an expected 1.6 genres and 25.6% of pairs). Between persons, $\tau_{between} = 0.9$ (95% CI [0.06, 2.13]; an expected 3.5 genres and 36.6% of pairs), $\sigma_{s,between} = 0.19$ (95% CI [0.01, 0.65]; an expected 0.14 genres and 2.7% of pairs), and M2-D's between-person raw SD was 2.42 (95% CI [1.49, 3.54]; an expected 9.3 genres and 75.9% of pairs), again inflated by estimation error.

Robustness

Two sensitivity analyses tested whether the conclusions depended on analytic choices. Neither changed the H1 conclusion. Adjusting for age, gender, and ethnicity shifted the H1 coefficients by less than 0.01 points and changed no ROPE verdict (Appendix D, Table A0.2). Refitting on multiply imputed data (m = 100) left neither H1 coefficient practically significant under the preregistered 90% CI grid (equivalence tests are two one-sided tests at $\alpha = .05$, which correspond to a 90% rather than a 95% interval; Lakens (2017)) but, unlike the complete-case analysis, rejected genre-coefficient equality in neither pooled dimension (Appendix D, Table A0.4). Appendix D gives full specifications, results, and diagnostics for both analyses, together with the predictive model comparison and the preregistered frequentist analyses (Table A0.1): no H1 coefficient was practically significant under the preregistered grid with or without mobile play, and the within-person joint test for H2 rejected equality under full but not under primary genre attribution.

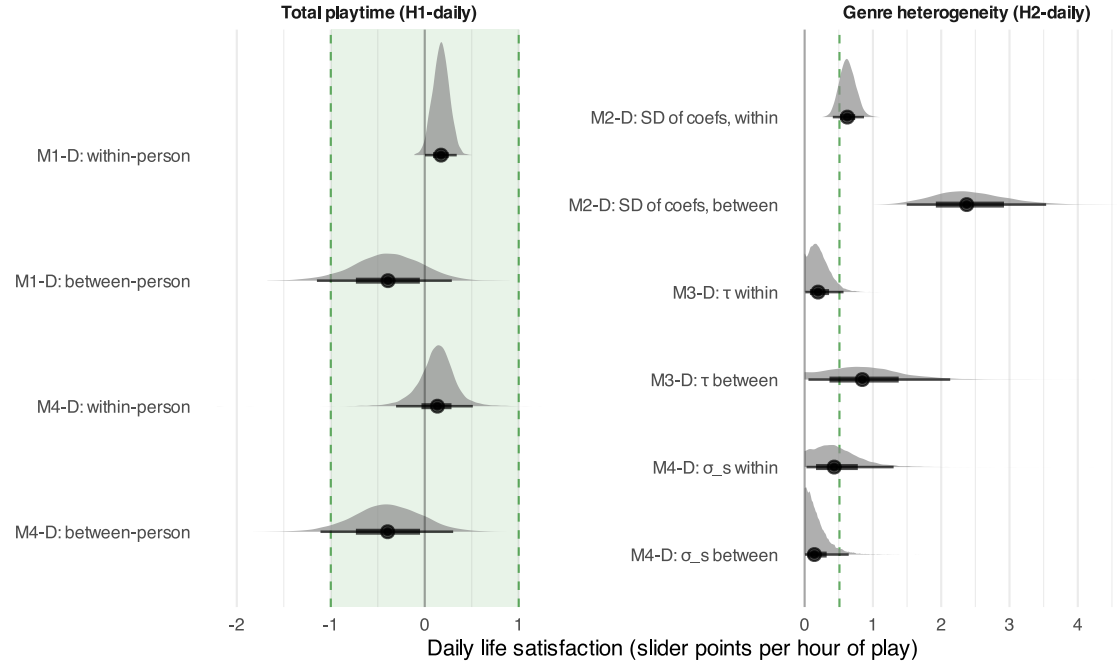


Figure 6. Exploratory Daily-Timescale Posteriors (Life Satisfaction). Left: total-playtime effects from the standalone H1 model (M1-D) and the multi-membership model's fixed effects (M4-D). Right: genre heterogeneity from all three specifications: M2-D (SD across the 14 independently estimated coefficients, computed at each posterior draw), M3-D (hierarchical-prior τ), and M4-D (multi-membership σ_s). Green band: ROPE (±1 slider point per hour) for the total-playtime effects. Dashed line in the right panel: reference value of 0.51 ($1 \div 1.96$), the spread at which 95% of genre effects would lie within one point of the average genre effect; the value for 95% of genre pairs differing by less than one point is 0.36. Both are reference points, not decision criteria (Methods). Thick: 66% credible interval; thin: 95% CI.

Discussion

When previous work has reported that time spent gaming carries no meaningful association with wellbeing, a common objection is that surely it must depend on the type of game one plays. After all, a relaxing puzzle game and a competitive shooter should not plausibly have the same effect on the person playing them. Video game heterogeneity is the standing explanation for why aggregate playtime nulls should not be taken at face value: playtime is a poorly specified exposure whose associations average over distinct causal contrasts (Ballou, Hakman, et al., 2025; Magnusson et al., 2024), and a null total association can in principle be produced by heterogeneous true effects that offset in aggregate (Yarkoni, 2022). Genre is the most widely used name for those contrasts (Starosta et al., 2024), which makes it the natural first place to look.

Continued debate about how different games might differ in their effects on wellbeing persists in part because of measurement challenges, whereby prior genre research has been subject to at least one of three limitations, and usually multiple. First, genre assignment is inconsistent and unreliable: labels are typically self-assigned by participants or researchers, the same games are classified into different genres across studies, and nearly all taxonomies force each game into a single genre despite modern games being multi-genre by design (King et al., 2019; Starosta et al., 2024). Second, playtime has often been self-reported rather than logged, yet self-reported and logged media use correlate only moderately, with a meta-analytic estimate of $r = 0.38$ and fewer than one in ten self-reports falling within 5% of the logged value (Parry et al., 2021). Our own positive control, which compared self-reported biweekly playtime with the logged total, gave $r = 0.481$: somewhat above the meta-analytic value, but still a correlation at which self-reports share only 23% of their variance with logged play, and one that may be depressed because the self-reports can also cover play on platforms outside our telemetry. Self-report and telemetry therefore agree on who plays more and who plays less, but too loosely to stand in for one another. Third, studies often cover a single game or a handful of titles (Brühlmann et al., 2020; Johannes et al., 2021; Larrieu et al., 2023), leaving genre contrasts to be assembled across different studies that classified and measured both play and genre incommensurably, and often years apart in a medium whose genre labels drift as it evolves (Cohen, 1986; Vargas-Iglesias, 2020), so that a label such as adventure need not have covered the same games in each.

This study represents an advancement on all three fronts. Genre labels follow a single crowd-sourced taxonomy applied uniformly to every title across all platforms, retaining multiple genre labels where relevant. Playtime is measured as objective session-level telemetry across Nintendo, Xbox, and Steam rather than self-reported. And all 14 genres are estimated jointly within a single sample, so genre contrasts emerge from one dataset with a common measurement of play, genre, and wellbeing.

The playtime-wellbeing association is practically equivalent to zero

When participants played more than their own average in a given two-week wave, their wellbeing barely changed, consistent with previous digital trace studies that have repeatedly found aggregate playtime to be a poor predictor of wellbeing fluctuations (Ballou, Sewall, et al., 2024; Ballou, Vuorre, et al., 2025; Larrieu et al., 2023). Between participants, practical equivalence was likewise established.

Inverting the estimates shows what these associations would amount to even if they were causal. Within persons, wellbeing would shift by one minimally important difference only after roughly 75 additional hours of play per day at the point estimate, and after 20 additional hours at the credible bound most favourable to an association, more extreme than the 10 additional daily hours that Vuorre et al. (2022) derived from game-level telemetry. Between participants, the habitual difference needed to reach the same threshold lies between 9 and 19 hours of daily play. Both requirements exceed the roughly five hours of leisure that US adults have in a day (Sturm & Cohen, 2019), and changes of this order do not occur in practice: of the 1,442 participants observed across adjacent waves, only 113 ever changed their average daily play by 4 or more hours from one wave to the next. An association that no achievable play exposure can reach is, for every practical purpose, indistinguishable from no association at all.

The contrast between light and heavy players deserves one further remark. Participants who habitually play more reported slightly lower average wellbeing, yet the same participants' wellbeing did not fall in waves when they played more than usual. This within-null, between-negative pattern recurs across observational media research and is most plausibly a selection effect, in which people with lower baseline wellbeing spend more time in games rather than games lowering wellbeing (Ballou, Vuorre, et al., 2025; Kievit et al., 2013; Robinson, 1950; Rohrer & Murayama, 2023), although a cumulative effect accruing over years would look the same at these timescales (cf. Vuorre et al., 2022). Whichever explanation holds, the difference is too small to matter, so public claims about gamers' mental health are, in effect, reports about who plays, not about what playing does.

The causal estimates from Japanese console lotteries point the other way, at 0.2 to 0.5 standard deviations per additional daily hour among lottery winners who acquired a console because they won (Egami et al., 2024; Egami et al., 2026). Those are local effects of acquiring a new console, measured with self-reported play over the preceding month and diminishing at longer play durations, whereas our within-person estimates describe logged deviations from each person's usual playtime among people who already play. The two estimands need not agree: gaining access to a console and playing an extra hour on a console one already owns are different versions of the same nominal exposure (Magnusson et al., 2024).

Associations are close to zero in every genre

We expected genres to differ in how playtime relates to fluctuations in general mental wellbeing. The preregistered joint test supports this prediction within persons, where the 14 genre coefficients are not all equal, but not between persons, and under the preregistered primary-genre attribution neither test rejects equality (Appendix D). The differences it detects lie between genre effects that are all close to zero: under the hierarchical prior, the 14 within-person genre estimates span only −0.019 to +0.017, with no estimate more than a third of the SESOI from zero, and the most-played genres by total playtime (Role-playing, Adventure, Strategy, Shooter; see Figure 2) sit inside this same narrow band. The spread estimated by the same model implies an expected 0.08 of the 14 genres more than 0.06 from the average genre effect and 2.6% of genre pairs more than 0.06 apart, and on average 0.7% of its within-person and 4.2% of its between-person genre-effect distribution lie outside the ROPE. The multi-membership model is less precise than the hierarchical prior model: its fitted spread implies that about 14% of genres (95% CI [0%, 51%]) deviate from the average slope by more than 0.06, an expected 7.1 of the 14 genres at the upper credible bound, so differences confined to a few genres cannot be excluded. Inference about single genres is limited as well, because the credible intervals for rarely played genres remain wide: under the preregistered model, the widest within-person interval (Point-and-Click) is about 5 times wider than the narrowest. Genres were statistically distinguishable only within persons and only when each game counted towards all its genres, and under the hierarchical prior no genre's estimated association approached the SESOI; any defence of playtime nulls that appeals to the heterogeneous nature of video games must look to partitions finer than genre.

This result does not contradict the genre differences reported before, because those concern other outcomes and other designs. The most consistent findings involve gaming disorder symptoms, where MMORPG, shooter, and strategy play stand out in cross-sectional surveys of self-reported genre preference (Elliott et al., 2012; King et al., 2019; Na et al., 2017; Rehbein et al., 2021), and player experience, where genres differ in immersion, autonomy, and frustration (Hazel et al., 2022; Johnson et al., 2015). Genre also explains little of why people play: among more than 14,000 highly engaged players, correlations between six gaming motives and genre preference ranged from −0.25 to 0.20 (Király et al., 2022). Our estimand, the association between logged exposure to a genre and general wellbeing, is different, and on that estimand every genre's association lies close to zero.

A Susceptible Minority Is Possible but Small

Still, a null average could conceal a minority for whom play, in total or in specific genres, genuinely matters (cf. Karhulahti et al., 2025). The random slope in M1 addresses the total-playtime case by modelling each participant's own within-person association between playtime and wellbeing. The relevant quantity is the share of participants the model implies lie beyond the SESOI: combining the posterior for the average within-person effect with the posterior for the person-level slope standard deviation (0.029, 95% CI [0.003, 0.058]) under the model's normality assumption, an estimated 8.1% of participants (95% CI [0%, 30.2%]) have a true within-person association exceeding 0.06 SWEMWBS points per daily hour in either direction. A susceptible minority is therefore entirely compatible with our results; what the data constrain is that such susceptibility is neither common nor, for most of the distribution, large at these timescales. Person-specific slope estimates are heavily shrunk towards the mean given the number of waves per participant, so the model quantifies how heavy the tails are without being able to identify which individuals occupy them; targeted designs with denser within-person sampling would be needed for that.

Shortening the Window Does Not Recover an Effect

A further possibility is temporal, namely that the wellbeing effects of play might surface and dissipate faster than a biweekly retrospective survey can register. To test this claim, the exploratory daily analysis shortens the window from fourteen days to one and reproduces the total-playtime conclusion. Within persons, every total-playtime estimate lies inside the equivalence region under every specification, and the same-day point estimate is positive, about a quarter larger than its biweekly counterpart in within-person standard-deviation units (0.014 versus 0.011 s_w per hour). The direction is consistent with the session-level affective uplift reported among *PowerWash Simulator* players (Vuorre, Ballou, et al., 2024), though its resolution is fragile across the robustness refits and should be read as suggestive. Its practical magnitude is negligible either way: shifting daily life satisfaction by one minimally important difference (5 points on the 0–100 slider, the daily counterpart of the 0.3-point SWEMWBS difference behind the preregistered SESOI) would require 29 additional hours of same-day play, and under the hierarchical prior the daily within-person genre spread implies an expected 0.10 of the 14 genres more than one slider point per hour from the common genre effect. Therefore, moving the measurement closer in time, on its own, does not conjure a meaningful effect.

Are all games the same?

The question this study set out to answer rests on an undisputed fact, that games differ psychologically, and although current genre labels are not an effective way to operationalise those differences when it comes to wellbeing outcomes, the general intuition persists. Game scholars have argued for decades that genre labels are conventions of marketing and library organisation rather than coherent descriptions of play: they conflate gameplay style, purpose, and mood in a single word (Lee et al., 2014), fail elementary classificatory tests (Clarke et al., 2017), and drift as the medium evolves (Cohen, 1986; Vargas-Iglesias, 2020). Even the literature most committed to genre-specific effects has reached the same conclusion: the researchers who documented attentional benefits of action games now argue that industry genre labels have

become poor proxies for the mechanics that produce those benefits, and call for a taxonomy of the processes engaged during play instead (Bavelier & Green, 2019; Nguyen & Bavelier, 2023). Games within one genre can differ sharply in structure, as Call of Duty and Halo do within the shooter genre (Saini & Hodgins, 2023). The pathways plausibly connecting play to wellbeing seem to operate at levels that genre labels cannot index (Ballou, Hakman, et al., 2025). For example, need satisfaction depends on autonomy-supportive design features (Ballou et al., 2023; Peng et al., 2012), social capital depends on co-play relationships (Depping et al., 2018; Vella et al., 2015), and affect regulation depends on the match between gameplay and current emotional state (Kosa et al., 2020).

The broader point is that, as many have argued theoretically but have lacked empirical data to show conclusively, other aspects of game play besides genre are likely more meaningful for wellbeing. Structural characteristics (e.g., avatar customisability) are one such aspect, and dedicated taxonomies now exist for characterising these (e.g., Saini & Hodgins, 2023). In-game behaviour is another: within a single game, behavioural variation in how people play is linked to meaningful wellbeing differences even when genre is held constant (e.g., Hakman et al., 2026). How players fit gaming into their lives is a third: among casual Nintendo players, self-rated gaming life fit predicted wellbeing an order of magnitude more strongly than any playtime measure (Ballou, Vuorre, et al., 2025). Community-assigned tags that describe mechanics, themes, and social structure may offer a practical middle ground between genre labels and title-level coding (Li, 2020; Starosta et al., 2024).

Our data leave two parts of this picture untested. Mobile games were recorded only as daily minutes without titles; they enter only the first positive control and the preregistered all-platform H1 test in Appendix D, which likewise finds no practically significant association, and none of the genre analyses (see Limitations). Monetisation is the other: free-to-play models, loot boxes, and the related design features that have been argued to harm players (Rehbein et al., 2021; Saini & Hodgins, 2023) need not be spread evenly across genres, so a genre contrast may partly carry a monetisation contrast; in one clinical sample the two were entangled closely enough that the authors proposed monetisation model as the more tractable partition (André et al., 2024).

Methodological Contribution: A Model for Overlapping Categories

Methodologically, the multi-membership model transfers beyond games. To our knowledge this is the first application in media psychology of the model class, developed originally for educational data where students attend multiple schools (Cafri et al., 2015; Fielding & Goldstein, 2006). Any media effects question with overlapping content labels has the same structure (e.g., streaming titles with multiple genre tags, news diets spanning outlets with overlapping orientations, social media feeds spanning topic categories) and can be answered with the same model. Our code is freely available to support future applications of this method.

Limitations

Several limitations constrain these conclusions and point at the work that should come next. First, the design is observational. The within-person decomposition removes stable confounds but not time-varying ones, and players themselves name illness, grief, work pressure, and caretaking as events that move both play and wellbeing, sometimes in opposite directions (Ballou, Földes, Hakman, et al., 2025); our causal statements are correspondingly guarded (Rohrer & Murayama, 2023). Second, the SWEMWBS is a global instrument assessed over a two-week recall window. Gaming sits alongside factors like sleep, work, and social life for wellbeing variance, and domain-specific outcomes (affect, sleep, connectedness) could show associations a global measure averages away. The exploratory daily analysis takes one step down this temporal hierarchy, but future work should consider pairing session-level telemetry with momentary assessments and using time-use data to partial gaming from competing activities. Third, coverage is imperfect: some participants used additional platforms whose telemetry we could not capture, Nintendo telemetry covers first-party titles only (Ballou, Földes, Vuorre, et al., 2025), the main and genre analyses excluded mobile play, which was recorded only as daily minutes without titles or metadata and enters only the first positive control and the preregistered all-platform H1 test in Appendix D, and in-study minutes are dominated by Steam (84%, versus 12.5% Xbox and 3.5% Nintendo), so the genre findings are predominantly findings about PC play by UK and US adults. Fourth, participants were adults aged 18 to 40 at screening, so our conclusions say nothing about children or adolescents, for whom both play patterns and susceptibility may differ. Nor do they extend to disordered play in clinical or at-risk populations, where genre-linked structural characteristics remain plausibly relevant (André et al., 2024; Rehbein et al., 2021). Stated as constraints on generality (Simons et al., 2017), our conclusions apply to adults aged 18 to 40 in the UK and US, to console and PC play recorded during the Open Play study period, and to the IGDB genre taxonomy; we have no reason to believe they depend on other characteristics of the sample, but they have not been tested beyond it.

Data and Code Availability

The Open Play dataset (v1.3.0) is openly available on Zenodo at https://zenodo.org/records/23419571, released under a CC0 licence with a supplemental no-reidentification clause (Ballou, Földes, Vuorre, et al., 2025; Ballou et al., 2026). All analysis code is available at https://github.com/Thomhak/open-play-genres. Preregistration materials for the Stage 1 programmatic Registered Report are available at https://osf.io/mvngt/.

Funding

This research was supported by the UK Economic and Social Research Council (ES/W012626/1 and ES/

Y010736/1). AKP was supported by the Huo Family Foundation.

Conflicts of Interest

The authors declare no competing financial, intellectual, or institutional interests relevant to this research. The disclosures below are provided in the interest of full transparency.

All members of the research team have worked on scientific projects funded by government and charitable grants that used data from tech companies provided under a data-sharing agreement. No authors have received funding or compensation from the tech companies or other industries connected to the work presented here.

MV has provided in-kind consultations and served as a nonpaid panel member for Meta and K-Games. AKP has served in advisory or expert roles for the Sync Digital Wellbeing Program (2022–2024), the Google Expert Advisory on Youth and Technology (2025), the OpenAI Expert Council on Well-Being and AI (2025), and UK Department for Science, Innovation and Technology–funded research on children's smartphone and social media use (University of Cambridge, 2025). He is a member of the UK Department for Culture, Media & Sport College of Experts. Any industry fees are directly donated to charity, and his research is conducted in accordance with University of Oxford academic integrity policies.

Data-sharing partners had no role in the design, analysis, or publication of results.

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A1 Appendix A: Deviations from Preregistration

This appendix lists all deviations from the Stage 1 preregistration (Ballou, Földes, et al., 2024) in the structured disclosure format of Willroth and Atherton (2024): the original wording, what was done instead, the reason, the timing relative to data access and results, and the impact on interpretation. Analyses that were not preregistered (M3, M4, and the daily-timescale analysis) are not deviations and are labelled exploratory where they appear. Changes made upstream during data collection, outside the authors' control, are consolidated as D6.

#	Topic	Type	Timing
D1	Genre assignment by full attribution	Data preparation	After data access, before results known
D2	Genre taxonomy	Data preparation	After data access, before results known
D3	Bayesian estimation and priors	Inferential criteria	After data access, before results known
D4	H1 platform scope without mobile	Variables	After data access, before results known
D5	Missing-data protocol	Analytic approach	After complete-case results known
D6	Upstream data-collection changes	Data collection procedures	During data collection, before data access

A1.1 Structured disclosures

A1.1.1 D1: Genre assignment by full attribution

Field	Content
Type	Data preparation
Original wording	Games are classified by their primary (first) IGDB genre.
Deviation description	A game's full playtime counts towards every one of its IGDB genre labels (up to 16 per game). Genre hours are therefore exposures to games carrying each label and are not mutually exclusive; total playtime for H1 is computed separately from raw sessions.
Reason	Most games carry several labels; using only the first discards classification information and depends on IGDB's label order.
Timing	After data access, before results known
Reader impact	Each genre coefficient estimates the association with exposure to games carrying that label, not the effect of a genre in isolation. Under the preregistered primary-genre attribution, the within-person joint Wald test does not reject the equality of the genre coefficients ($p = 0.216$), whereas it does under full attribution ($p = 0.025$), so the support for H2a depends on this choice (Appendix D, Table A1.1, primary-attribution rows).

A1.1.2 D2: Genre taxonomy

Field	Content
Type	Data preparation
Original wording	IGDB sub-genres are collapsed as follows: Hack and slash/Beat 'em up → Action; Point-and-click and Visual Novel → Adventure; Turn-based strategy, Real-time strategy, and Tactical → Strategy; Quiz/Trivia → Puzzle.

Field	Content
Deviation description	Hack-and-slash/Beat-‘em-up, Point-and-click, and Visual Novel are kept as separate categories; MOBA joins Strategy, Card & Board Game joins Puzzle, Pinball and Music join Arcade, and Indie is dropped as a non-substantive label, giving 14 categories (Appendix B).
Reason	The preregistered collapsing merged genres that prior literature treats as psychologically distinct.
Timing	After data access, before results known
Reader impact	Per-genre estimates refer to different category boundaries than the preregistration implies. The retained sub-genres are rarely played and have wide credible intervals.

A1.1.3 D3: Bayesian estimation and priors

Field	Content
Type	Inferential criteria
Original wording	H1: equivalence test (TOST) with SESOI = ±0.06 SWEMWBS points per hour/day, using 90% CIs. H2: joint Wald test via `car::linearHypothesis()`. No priors are specified (all models frequentist).
Deviation description	M1 and M2 are estimated as Bayesian multilevel models with `brms` (Bürkner, 2017) rather than with `lmer`. H1 is evaluated against the preregistered ROPE [−0.06, +0.06], with a 95% credible interval entirely inside the ROPE establishing practical equivalence (Kruschke, 2018), in place of the preregistered decision grid for the 90% confidence interval (Methods; Appendix D). The preregistered joint Wald test for H2 is retained on the frequentist counterpart of M2 and answers H2; the size of any departure is described, without a verdict, by the posterior SD of M2′s genre coefficients and the shares of genres and genre pairs it implies beyond the SESOI (Methods). All priors are the `brms` defaults: improper flat priors on the coefficients, a Student-*t*(3, 3.3, 2.5) prior on the intercept, centred at the sample median of SWEMWBS, and half-Student-*t*(3, 0, 2.5) priors on the standard deviations (Appendix F). An earlier version used weakly informative priors; the switch to the defaults was made after those results were known.
Reason	Alongside the decision, the ROPE gives a continuous measure of practical equivalence (the posterior share inside it). With flat priors on the slopes, the data alone determine their estimates, so the Bayesian estimates are directly comparable with the preregistered maximum-likelihood analysis.
Timing	After data access, before results known (Bayesian estimation); the switch to the `brms` default priors was made after the results of the earlier priors were known
Reader impact	The SESOI and the equivalence region are unchanged, so the risk of bias is low. All priors are the `brms` defaults (Appendix F), so no analyst-chosen prior scale enters the models. The preregistered frequentist analyses, with the equivalence tests on 90% confidence intervals and the joint Wald tests, are reported in full in Appendix D (Preregistered Frequentist Analysis, Table A1.1).

A1.1.4 D4: H1 platform scope without mobile

Field	Content
Type	Variables
Original wording	H1 uses total playtime across all five platforms (Nintendo, Xbox, Steam, iOS, Android).

Field	Content
Deviation description	H1 total playtime uses Nintendo, Xbox, and Steam only. Mobile platforms provide daily minutes without game titles (D6), so including them for H1 but not for H2 would give “total playtime” two different meanings.
Reason	Consistency between H1 and H2.
Timing	After data access, before results known
Reader impact	Mobile play is omitted from the H1 exposure; it made up 3.4% of all-platform playtime in the analysed person-waves, and the preregistered all-platform H1 model likewise finds no practically significant association in either dimension (Appendix D, Table A1.1, all-platform rows). The findings do not speak to mobile play.

A1.1.5 D5: Missing-data protocol

Field	Content
Type	Analytic approach
Original wording	Multiple imputation via `mice` (predictive mean matching, wide format); report both complete-case and imputed results.
Deviation description	All outcome missingness is wave nonresponse: 3,490 of 10,008 person-waves (34.9%) have no SWEMWBS. We therefore imputed on the full participant × wave grid with two-level MICE (`2l.pmm`, participants as clusters; m = 100 by von Hippel’s rule) and report pooled frequentist results alongside the complete-case Bayesian models (Appendix D).
Reason	Single-level wide-format PMM ignores the hierarchical structure and cannot address wave nonresponse.
Timing	Adopted after the complete-case results were known; the protocol was taken over unchanged from the sibling Stage 2 studies of the same programmatic report rather than tailored to these results
Reader impact	The H1 conclusions are unchanged under imputation, whereas the pooled within-person joint test for H2 does not reject the equality of the genre coefficients, unlike the complete-case test; the pooled between-person H1 interval widens because person means are uncertain for participants with few observed waves. Agreement between complete-case and imputed H1 results does not rule out bias from nonresponse that depends on unobserved wellbeing.

A1.1.6 D6: Upstream data-collection changes

Field	Content
Type	Data collection procedures
Original wording	Age range 18–75 (UK) and 18–30 (US); at least 75% of gaming time on included platforms; Android telemetry as session-level ActivityWatch exports.
Deviation description	Made by the Open Play team before this study’s data access: screening admitted ages 18–40 in both regions, the coverage threshold was lowered to 50%, and Android data were collected as daily minutes from Digital Wellbeing screenshots. Reported in full by Ballou, Földes, Vuorre, et al. (2025).
Reason	Recruitment and data-collection constraints upstream; outside the authors’ control.
Timing	During data collection, before data access
Reader impact	Findings generalise to adults aged 18–40; the lower coverage threshold adds measurement error to total playtime; Android contributes daily totals only (see D4).

A2 Appendix B: Genre Mapping

Genre labels from all three platforms resolve to the study taxonomy in two steps: Xbox store labels are first mapped onto IGDB genres (per the preregistration's Appendix B), and raw IGDB labels (23 distinct labels observed in the study metadata) are then collapsed into 14 analysis-level categories. Table A2.1 presents both steps in one place: each row is an implemented study genre, with the raw IGDB labels it subsumes, the Xbox store labels routed to it, and, where the preregistered mapping (which would yield 17 distinct genres on the observed labels) treated a label differently, the preregistered assignment. The implemented mapping diverges from the preregistered one in both directions: it preserves granularity for sub-genres that prior literature treats as distinct (Hack-and-slash/Beat-'em-up, Point-and-click, Visual Novel), collapses more aggressively elsewhere (MOBA into Strategy, Card & Board Game into Puzzle, Pinball and Music into Arcade), and excludes Indie as a non-substantive label (see Appendix A, Deviation D2). Two details sit outside the table: the Xbox label Family + Kids has no IGDB equivalent and is resolved by the title's secondary genre, and IGDB includes no top-level "Action" category, so what is colloquially called "action" is distributed across Shooter, Hack-and-slash, Fighting, and Platform.

Study genre	IGDB label(s) subsumed	Xbox store label(s)	Preregistered assignment (where different)
Adventure	Adventure	Action + Adventure	Also subsumes Point-and-click and Visual Novel
Arcade	Arcade, Pinball, Music	Classics, Music	Pinball and Music retained as separate genres
Fighting	Fighting	Fighting	
Hack-and-slash/Beat-'em-up	Hack and slash/Beat 'em up		Collapsed into Action
Platform	Platform	Platformer	
Point-and-click	Point-and-click		Collapsed into Adventure
Puzzle	Puzzle, Quiz/Trivia, Card & Board Game	Puzzle + Trivia, Card + Board, Casino	Card & Board Game retained as a separate genre
Racing	Racing	Racing + Flying	
Role-playing	Role-playing (RPG)	Role Playing	
Shooter	Shooter	Shooter	
Simulator	Simulator	Simulation	
Sport	Sport	Sports	
Strategy	Strategy, Turn-based strategy (TBS), Real time strategy (RTS), Tactical, MOBA	Strategy, Multi-Player Online Battle Arena	MOBA retained as a separate genre
Visual Novel	Visual Novel		Collapsed into Adventure
(excluded)	Indie		Retained as Indie

Table A2.1. Mapping from platform genre labels to the 14 study genres. Xbox store labels map onto IGDB genres, which are then collapsed into the study taxonomy. The final column gives the preregistered assignment where it differs; blank cells indicate identical treatment.

A3 Appendix C: Sample and Genre Descriptive Statistics

A3.1 Participant Demographics

Characteristic	Total	US	UK
N	**1668**	**967**	**701**
Age (years)	27 (5.1)	27 (5.1)	26.9 (5.1)
Gender			
Man	1088 (65.2%)	602 (62.3%)	486 (69.3%)
Woman	476 (28.5%)	292 (30.2%)	184 (26.2%)
Non-binary/Other	100 (6%)	71 (7.3%)	29 (4.1%)
Ethnicity			
White	1165 (69.8%)	585 (60.5%)	580 (82.7%)
Asian	138 (8.3%)	86 (8.9%)	52 (7.4%)
Black	109 (6.5%)	94 (9.7%)	15 (2.1%)
Mixed/Multiple	165 (9.9%)	138 (14.3%)	27 (3.9%)
Other	47 (2.8%)	43 (4.4%)	4 (0.6%)
Platform			
Nintendo	523 (31.4%)	326 (33.7%)	197 (28.1%)
Steam	1371 (82.2%)	762 (78.8%)	609 (86.9%)
Xbox	163 (9.8%)	163 (16.9%)	0 (0.0%)
iOS	93 (5.6%)	66 (6.8%)	27 (3.9%)
Android	73 (4.4%)	50 (5.2%)	23 (3.3%)

Values are M (SD) or N (percent). Platform counts are non-exclusive (participants may have data from multiple platforms).

Table A3.1. Participant characteristics by country.

A3.2 Genre Playtime Descriptives

Genre	n games	Total hours	Mean min/ day	Top 5 titles
Role-playing	1013	88696	58	Baldur's Gate III [4.3%], Stardew Valley [3%], Elden Ring: Nightreign [2.9%], Final Fantasy XIV Online [2.8%], Umamusume: Pretty Derby [2.5%]
Adventure	1756	87041	57	Stardew Valley [3%], Path of Exile [2.6%], Warframe [2.2%], Peak [2.1%], The Binding of Isaac: Rebirth [1.7%]
Strategy	960	75006	49	Counter-Strike 2 [6.6%], Baldur's Gate III [5%], Dead by Daylight [4.8%], Balatro [3.7%], Stardew Valley [3.5%]
Shooter	692	60931	40	Marvel Rivals [12%], Counter-Strike 2 [8.1%], Overwatch 2 [3.9%], Warframe [3.2%], Helldivers 2 [3.2%]
Simulator	1054	57766	38	Stardew Valley [4.6%], RimWorld [4%], Umamusume: Pretty Derby [3.9%], Schedule I [3.4%], Football Manager 2024 [2.2%]
Puzzle	572	18144	12	Balatro [15.3%], The Binding of Isaac: Rebirth [8.1%], Slay the Spire [4.5%], Magic: The Gathering Arena [4.5%], The Legend of Zelda: Tears of the Kingdom [4.5%]
Sport	189	13260	9	Umamusume: Pretty Derby [17%], Rocket League [13.3%], Football Manager 2024 [9.4%], C3C63F46… [4.6%], Rematch [3.9%]
Hack & Slash / Beat 'em Up	160	10989	7	Path of Exile [20.3%], Monster Hunter Wilds [12.7%], Palworld [11.7%], Path of Exile 2 [7.9%], Torchlight: Infinite [3.4%]
Platform	310	11725	8	Peak [15.9%], MapleStory [11.4%], Terraria [10%], Super Smash Bros. Ultimate [7%], Splatoon 3 [5.8%]
Arcade	241	10016	7	Umamusume: Pretty Derby [22.5%], The Binding of Isaac: Rebirth [14.7%], Deltarune [7.4%], Mario Kart 8 Deluxe [4.8%], Spacewar [4.3%]
Racing	146	7544	5	Rocket League [23.4%], F-Zero DSX [11.7%], Blender Bros. [7.4%], Mario Kart 8 Deluxe [6.4%], Grand Theft Auto V Enhanced [5.6%]
Fighting	145	5029	3	Super Smash Bros. Ultimate [16.2%], WWE 2K25 [7.6%], Tekken 8 [5.7%], Naval Action [5.6%], Brawlhalla [5.5%]
Visual Novel	103	4346	3	Umamusume: Pretty Derby [51.7%], Blue Archive [13.5%], Limbus Company [7.7%], Date Everything! [6.2%], The Hundred Line: Last Defense Academy [4.3%]
Point-and-Click	108	1461	1	Spirit City: Lofi Sessions [49.3%], The Roottrees Are Dead [5.4%], Escape Simulator [3.2%], Danganronpa V3: Killing Harmony [3.2%], The Great Ace Attorney Chronicles [2.7%]
Total (unique, deduplicated)	3651	222125	105	

Table A3.2. Playtime descriptives by genre. A game's playtime is attributed in full to each of its IGDB labels, so rows are not mutually exclusive and n games exceeds the total unique titles. n games = unique titles; Total hours = playtime attributed to that genre; Mean min/day = average minutes per person-day; Top 5 titles = most-played games by total minutes, with share of genre playtime in brackets (Xbox titles anonymised; their identifiers are shortened to the first eight characters, followed by an ellipsis). The Total row uses deduplicated playtime and distinct titles.

A4 Appendix D: Sensitivity and Supplementary Analyses

This appendix first reports the preregistered frequentist analyses on the analytic sample and under two preregistered specifications. It then reports the two sensitivity analyses summarised under Robustness in the main text: demographic covariate adjustment and multiple imputation of wave nonresponse (with distributional diagnostics). The predictive model comparison (whether genre terms improve out-of-sample prediction) follows, and finally the design details and sensitivity refits of the daily-timescale analysis.

A4.1 Preregistered Frequentist Analysis

The preregistered analyses were frequentist: H1 was to be tested by an equivalence test on the 90% confidence intervals of the within-between model of total playtime, and H2 by joint Wald tests of the equality of the genre coefficients, both on multilevel models fitted with `lme4`. The Bayesian models in the main text are the models we interpret (Deviation D3), and the Results report the preregistered H2 tests on the analytic sample. This subsection reports all preregistered tests in one place (Table A4.1): on the analytic sample and under the two preregistered specifications from which the main analysis departs, total playtime across all five platforms for H1 (Deviation D4) and primary-genre attribution for H2 (Deviation D1). The all-platform total adds each wave's iOS and Android gaming minutes to its Nintendo, Xbox, and Steam playtime; 166 participants had mobile minutes in at least one analysed person-wave, and 9.0% of the analysed person-waves had any. Under primary attribution, each game's full playtime counts towards its first IGDB genre label that remains in the 14-genre taxonomy (Appendix B), so genre exposures are mutually exclusive. Each H1 model has a random intercept and a correlated random slope for within-person playtime, as preregistered, and was fitted by restricted maximum likelihood; each H2 model was fitted by maximum likelihood with a random intercept, because the preregistered uncorrelated random slopes gave a singular fit in both H2 analyses (table note). H1 verdicts follow the preregistered decision grid of the Stage 1 Design Table, applied to the lower (LB) and upper (UB) bounds of the 90% confidence interval: LB > 0.06 or UB < −0.06 indicates a practically significant positive or negative association; LB > 0 and UB > 0.06, or UB < 0 and LB < −0.06, a positive or negative association that may be practically significant; LB > 0 and UB < 0.06, or UB < 0 and LB > −0.06, a positive or negative association that is not practically significant; LB > −0.06 and UB < 0.06 with an interval that covers zero, no practically significant association; and an interval that covers zero and one or both bounds, an inconclusive result. The Design Table lists the absence of a practically significant association for every interval inside ±0.06, including those that exclude zero, and we applied the more specific classes to the latter; it also gives the first bound as LB > −0.06, which we read as LB > 0.06, the counterpart of UB < −0.06. H2 is supported if the joint test rejects equality at $\alpha = .05$.

Under the preregistered grid, both H1 coefficients on the analytic sample showed no practically significant association (within persons: 0.0037 [−0.0054, 0.0129]; between persons: −0.0153 [−0.0307, 0.0002]): each interval included zero and lay inside ±0.06. Mobile play made up 3.4% of all-platform playtime in the analysed person-waves. With it included, both verdicts were unchanged (between persons: −0.0148 [−0.0302, 0.0005]). None of the 4 H1 rows showed an association that is, or may be, practically significant. On the analytic sample under full attribution, the within-person joint test rejected the equality of the genre coefficients ($p = 0.025$) and the between-person test did not ($p = 0.315$), so H2a was supported and H2b was not. Under the preregistered primary-genre attribution, neither test rejected equality (within: $\chi^2(13) = 16.6$, $p = 0.216$; between: $\chi^2(13) = 15.4$, $p = 0.284$), so the support for H2a depended on the attribution rule. Primary attribution assigns each multi-genre game to a single genre, which leaves genres that are seldom listed first with few players: Visual Novel was the primary genre of games played by 6 participants, whereas 128 participants played games carrying the label. Its between-person coefficient had a standard error of 5.06, and the SD of the other 13 between-person coefficients was 0.094, so the between-person SD of 0.916 under primary attribution reflected the estimation error of this genre rather than a spread across genres.

A4.2 Demographic Covariates

The preregistration specified a sensitivity analysis adjusting the primary models for demographic covariates. We refit M1 (total playtime) and M2 (independent genre coefficients) with age, gender, and ethnicity added as population-level predictors, using the subset of participants with complete demographic data.

The adjusted models used 6,459 observations from 1,622 participants with complete demographic data. Table A4.2 compares the primary and covariate-adjusted estimates. The H1 coefficients shifted by at most 0.006 points after adjustment, and the ROPE conclusions were unchanged: >99.9% of the adjusted within-person posterior draws fell inside the ROPE (primary: >99.9%), and >99.9% of the adjusted between-person posterior draws (primary: >99.9%). One qualification: the adjusted between-person credible interval ([−0.039, −0.003]) excluded zero, unlike the unadjusted interval; this small negative association remained far inside the ROPE and is consistent with the selection account discussed under H1. The genre heterogeneity estimates were likewise essentially unaffected. Both adjusted models converged cleanly (max R-hat = 1.005, 0 divergent transitions). Demographic composition therefore did not account for the observed associations.

A4.3 Multiple Imputation

The preregistration specified multiple imputation by chained equations with predictive mean matching. We implemented the two-level grid protocol described in Methods (Missing Data): across the 10,008-cell participant × wave grid, 3,490 cells

Test	Specification	Participants	Person-waves	Estimate [90% confidence interval] or χ^2	p	Genre SD	Preregistered verdict
H1a	Analytic sample, console/PC	1,668	6,518	0.0037 [−0.0054, 0.0129]			No practically significant association
H1b	Analytic sample, console/PC	1,668	6,518	−0.0153 [−0.0307, 0.0002]			No practically significant association
H1a	All five platforms (D4)	1,668	6,518	0.0035 [−0.0053, 0.0123]			No practically significant association
H1b	All five platforms (D4)	1,668	6,518	−0.0148 [−0.0302, 0.0005]			No practically significant association
H2a	Analytic sample, full attribution	1,668	6,518	$\chi^2(13) = 24.8$	0.025	0.022	Equality rejected (supported)
H2b	Analytic sample, full attribution	1,668	6,518	$\chi^2(13) = 14.9$	0.315	0.057	Equality not rejected (not supported)
H2a	Primary attribution (D1)	1,668	6,518	$\chi^2(13) = 16.6$	0.216	0.044	Equality not rejected (not supported)
H2b	Primary attribution (D1)	1,668	6,518	$\chi^2(13) = 15.4$	0.284	0.916	Equality not rejected (not supported)

Analytic sample: qualified participants with at least one valid SWEMWBS (Open Play v1.3.0). H1 models: random intercept and correlated random slope for within-person playtime, fitted by restricted maximum likelihood. H2 models: random intercept, fitted by maximum likelihood; the preregistered structure with uncorrelated random slopes for the 14 within-person genre exposures is singular in both H2 analyses, with 11 (analytic sample, full attribution) and 9 (primary attribution) of the 14 slope SDs estimated at zero. Genre SD: SD of the 14 genre coefficient estimates, which includes their estimation error.

Table A4.1. Preregistered frequentist analyses on the analytic sample and under the preregistered specifications. H1 rows: within-person (H1a) and between-person (H1b) total-playtime coefficients in SWEMWBS points per hour/day with Wald 90% confidence intervals, classified by the preregistered decision grid (SESOI ±0.06; text above the table). H2 rows: joint Wald tests of the equality of the 14 within-person (H2a) and the 14 between-person (H2b) genre coefficients, with the SD of the 14 coefficient estimates as a descriptive summary.

(34.9%) had no observed SWEMWBS (Table A4.3), and m = 100 imputed datasets were generated. The pilot fractions of missing information for the H1 coefficients were 0.79 (within-person) and 0.82 (between-person): the between-person coefficient depends on person means that are genuinely uncertain for participants with few observed waves, and Rubin's rules propagate that uncertainty into the pooled intervals. Von Hippel's rule would require 137 imputations for a 5%-stable standard error; with m = 100 the achieved coefficient of variation of the pooled standard errors is approximately 0.06, which widens the pooled intervals slightly but does not affect the H1 verdicts.

Table A4.4 compares complete-case and pooled imputed results. The H1 conclusions were unchanged: both pooled 90% confidence intervals fell entirely within ±0.06, so under the preregistered grid neither coefficient showed a practically significant association on the imputed data. The between-person interval widened under imputation, reflecting that person-mean wellbeing is uncertain for participants with few observed waves.

For H2, the comparison separates the preregistered test from the size of the genre differences. As reported in the Results, the complete-case joint Wald test rejected *exact* equality of the 14 within-person genre coefficients ($\chi^2(13) = 24.8$, $p = 0.025$); under the preregistered decision rule this reading is limited to "at least one pair of genres differs", with

Quantity	Primary	Adjusted
H1a: Within-person total playtime	0.004 [−0.007, 0.015]	0.004 [−0.007, 0.015]
H1b: Between-person total playtime	−0.015 [−0.034, 0.003]	−0.021 [−0.039, −0.003]
H2a: Genre SD (within)	0.032 [0.020, 0.047]	0.032 [0.020, 0.048]
H2b: Genre SD (between)	0.082 [0.048, 0.129]	0.071 [0.041, 0.111]

Table A4.2. Primary versus demographic-adjusted estimates. H1 rows compare M1′s total-playtime coefficients with the same coefficients after adding age, gender, and ethnicity. H2 rows compare M2′s genre heterogeneity statistic (SD of the 14 genre coefficients, computed at each posterior draw) with its adjusted counterpart. Estimates in SWEMWBS item-mean points per hour/day.

Wave	Possible	Observed	Missing	% missing
1	1668	1668	0	0
2	1668	1163	505	30
3	1668	1086	582	35
4	1668	948	720	43
5	1668	870	798	48
6	1668	783	885	53

Table A4.3. SWEMWBS missingness by wave on the full participant × wave grid (analytic sample). Wave 1 was completed on the baseline day; later waves show progressive nonresponse.

Quantity	Complete-case	Imputed (pooled)	Verdict
H1a: Within-person total playtime	0.004 [−0.005, 0.013]	0.006 [−0.011, 0.023]	No practically significant association
H1b: Between-person total playtime	−0.015 [−0.031, 0.000]	−0.010 [−0.040, 0.020]	No practically significant association
H2a: Joint test, within-person genre coefficients equal	chi2(13) = 24.8, p = 0.025	F(13, 9239) = 0.63, p = 0.828	See text
H2b: Joint test, between-person genre coefficients equal	chi2(13) = 14.9, p = 0.315	F(13, 89721) = 1.04, p = 0.413	Not rejected (either)
Genre SD (within)	0.022	0.026	Descriptive
Genre SD (between)	0.057	0.057	Descriptive

Table A4.4. Complete-case versus multiply imputed estimates (m = 100; Rubin-pooled with Barnard-Rubin degrees of freedom). H1 rows: total-playtime coefficients from the same within-between model fitted by restricted maximum likelihood, with playtime centred over the observed waves of each participant (hence slightly different from the preregistered rows in Table A4.1, which centre over all survey waves), and verdicts from the preregistered decision grid for 90% confidence intervals (SESOI 0.06; Appendix D). H2 rows: joint tests of genre-coefficient equality (complete-case Wald chi-square; imputed pooled D1 F-test) and the SD of the 14 genre coefficients. The preregistered complete-case tests, with the preregistered specifications, are reported in Table A4.1.

no pairwise interpretation absent formal post-hoc tests. The SD of the 14 coefficient estimates, which describes the size of the differences and includes their estimation error, was 0.022 (complete-case) and 0.026 (imputed). On the imputed data the pooled D_1 tests rejected neither dimension (within: $p = 0.828$; between: $p = 0.413$), though these tests carried reduced power at this level of missing information (relative increase in variance 0.59 for the within-person constraints). Distributional diagnostics for the imputation are shown in Figure A4.1. In sum, accounting for wave nonresponse left the H1 conclusion intact, as neither total-playtime coefficient was practically significant, whereas whether the within-person joint test detected genre differences depended on the frame.

A4.3.1 Distributional Diagnostics

The central plausibility check for the imputation model is distributional: imputed wellbeing values should occupy the same range as the observed data and differ from it only in ways explicable by which participants are missing, not by model artefacts.

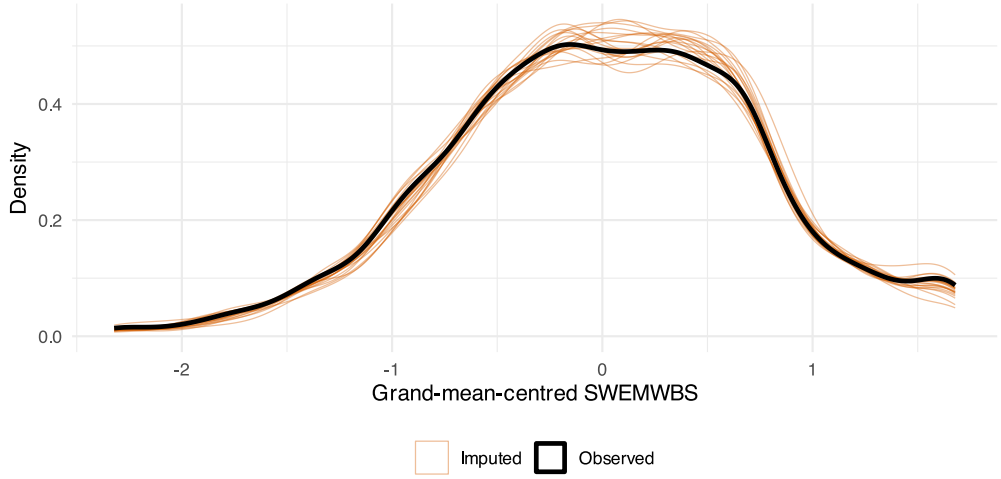


Figure A4.1. Multiple-Imputation Distributional Diagnostic. Observed grand-mean-centred SWEMWBS values (black) against imputed values in 20 of the 100 imputed datasets (thin orange); a subset is shown so the observed curve stays visible, and all 100 behave alike.

Figure A4.1 shows the imputed values to be distributionally almost indistinguishable from the observed data: pooled across all 100 imputations, the imputed values had mean 0.013 and SD 0.74 on the centred scale, against an observed mean of 0 (by construction) and SD of 0.75. The imputation model neither compressed the outcome's variance nor shifted nonresponders' imputed wellbeing to implausible regions of the scale. Convergence traces and strip plots are produced alongside the cached imputation (`output/qc_imputation/`).

A4.4 Predictive Model Comparison

A complementary way to ask whether genre matters is through predictive performance: does adding genre-specific terms to the total-playtime-only M1 baseline improve out-of-sample prediction? We assessed this using approximate leave-one-out cross-validation (LOO-CV) via Pareto-smoothed importance sampling (Vehtari et al., 2017). A positive difference in expected log predictive density (ELPD) relative to M1 indicates that genre terms help.

M1, the model with total playtime only, showed the highest expected log predictive density, and all three genre specifications predicted slightly worse out-of-sample (Table A4.5), with ELPD deficits between one and roughly two standard errors of the difference. Adding genre information, in any of the three specifications, therefore did not improve the prediction of wellbeing over total playtime alone. Pareto-k diagnostics flagged 4.1–4.3% of observations per model ($k > 0.7$; 265–281 observations), so the ELPD values are approximate; this affects the precision of the differences, not the ordering.

A4.5 Daily-Timescale Analysis: Design Details and Sensitivity Analyses

Local time was taken from each participant's reported timezone; Xbox sessions recorded with date-only timestamps (8% of sessions) had their minutes spread uniformly over the recorded day. Exclusions from the analysed daily sample were dominated by days without telemetry coverage; excluded responses entered an all-responses sensitivity refit that retains every response and scores unobserved play as zero. Between-person terms are habitual full-day hours, a participant's total telemetry over their observed diary period divided by days. The SESOI for the slider and the choice of its minimally important difference are described in Methods (Daily Timescale).

Four sensitivity refits accompany the exploratory daily analysis. Allowing first-order autocorrelated residuals with a within-person day trend (estimated AR(1) = 0.32) widened the within-person interval past zero (β = 0.13, 95% CI [−0.02, 0.29]), as did adjusting for window length (β = 0.14, 95% CI [−0.04, 0.31]). Excluding the 8% of Xbox sessions recorded with date-only timestamps, whose start times were floored to midnight and whose minutes were spread uniformly over

Model	dELPD	SE(d)	ELPD	SE	p_loo	LOOIC
M1: Total playtime only	0.0	0.0	−4197.5	77.9	1373.6	8395.1
M2: Independent coefficients	−1.0	6.1	−4198.6	77.7	1353.2	8397.1
M3: Hierarchical prior	−7.8	4.5	−4205.3	78.1	1359.5	8410.6
M4: Multi-membership	−8.2	3.4	−4205.7	78.3	1384.6	8411.5

Table A4.5. Leave-one-out cross-validation. Models ordered by expected log predictive density (ELPD); negative dELPD indicates worse out-of-sample prediction than the best-fitting model. SE(d) = standard error of the difference.

the recorded day, left the within-person interval above zero (β = 0.18, 95% CI [0.01, 0.35]). The all-responses refit, which retains every diary response and scores unobserved play as zero, gave the same between-person lean as the analysed sample (β = −0.36, 95% CI [−1.02, 0.28]). One qualification applies to the daily genre-level intervals: they inherit the exchangeable-residual assumption that the autocorrelation refit relaxes for the total-playtime model only. Like M3, the daily hierarchical-prior model was estimated in non-centred form; with the same sampler settings as M3 it produced 1 divergent transition, whereas the centred form had produced many in earlier fits, even at the stricter adapt_delta of 0.9995.

A5 Appendix E: Hypothesis Test Summaries

This appendix tabulates the preregistered joint tests for H2 and the posterior summaries and convergence diagnostics underlying the hypothesis tests and spread descriptors reported in the Results; the corresponding posterior distributions are shown in Figure 3 (H1) and Figure 4 (H2).

Hyp.	Model	Status	Parameter	Estimate	95% CI	% in ROPE	Genres > 0.06 (of 14)	Pairs > 0.06 (%)	Outside ROPE (%)	R-hat	Div.
H1: Total playtime, dose–response											
H1a	M1	Preregistered	Within-person slope	0.004	[−0.007, 0.015]	>99.9	–	–	–	1.006	0
H1b	M1	Preregistered	Between-person slope	−0.015	[−0.034, 0.003]	>99.9	–	–	–	1.006	0
H2a: Genre differences, within-person											
H2a	M2 (ML)	Preregistered	Joint Wald test: $\chi^2(13) = 24.8$, $p = 0.025$	–	–	–	–	–	–	–	–
H2a	M2	Preregistered	SD of genre coefficients	0.032	[0.020, 0.047]	–	0.98	18.5	7.9	1.004	0
H2a	M3	Exploratory	τ (hierarchical-prior scale)	0.016	[0.005, 0.031]	–	0.08	2.6	0.7	1.004	0
H2a	M4	Exploratory	σ_s (random-slope SD)	0.038	[0.004, 0.091]	–	2.0	24.8	16.3	1.003	0
H2b: Genre differences, between-person											
H2b	M2 (ML)	Preregistered	Joint Wald test: $\chi^2(13) = 14.9$, $p = 0.315$	–	–	–	–	–	–	–	–
H2b	M2	Preregistered	SD of genre coefficients	0.082	[0.048, 0.129]	–	6.2	58.4	45.4	1.004	0
H2b	M3	Exploratory	τ (hierarchical-prior scale)	0.019	[0.001, 0.054]	–	0.48	7.9	4.2	1.004	0
H2b	M4	Exploratory	σ_s (random-slope SD)	0.019	[0.002, 0.047]	–	0.35	6.9	4.1	1.003	0

All estimates are in SWEMWBS item-mean points per hour/day. ROPE = region of practical equivalence, −0.06 to +0.06, applied to the H1 slopes, which are the fixed effects of the standalone total-playtime model (M1), and, in the Outside ROPE column, to the H2 genre-effect distributions. H2 is answered by the preregistered joint Wald tests of equal genre coefficients, run on the maximum-likelihood fit of M2 with a random intercept (M2 (ML); Results). The three spread statistics describe genre-to-genre variation in exposure effects without a verdict: M2 reports the SD across its 14 independently estimated coefficients at each posterior draw (inflated by their estimation error), M3 the hierarchical-prior scale τ, and M4 σ_s, the SD of genre random-slope deviations under multi-membership weighting. Genres > 0.06: expected number of the 14 genres whose effect lies more than 0.06 from the average genre effect; Pairs > 0.06: share of genre pairs whose effects differ by more than 0.06; Outside ROPE: share of the genre-effect distribution, centred on the model's average genre effect, that lies outside ±0.06; all three are posterior means of per-draw quantities under normally distributed genre effects (Methods). R-hat is the maximum across all parameters of the fitted model; Div. = number of divergent transitions. Status follows the model: M1 and M2 are the preregistered models, under Bayesian estimation (Deviation D3), and M3 and M4 are exploratory; the spread statistics and their descriptors, including those of M2, were not preregistered.

Table A5.1. Hypothesis tests at the biweekly timescale, labelled by preregistration status: for H1, posterior estimates, 95% credible intervals and posterior mass inside the ROPE; for H2, the preregistered joint tests and the descriptive spread statistics with their three descriptors; and convergence diagnostics.

The exploratory daily-timescale analogue of this summary is given in Table A5.2.

Hypothesis	Model	Parameter	Estimate	95% CI	% in ROPE	Genres > 1 (of 14)	Pairs > 1 (%)	R-hat	Div.
H1a-daily	M1-D	Within-person slope	0.17	[0.00, 0.34]	>99.9	–	–	1.012	0
H1a-daily	M4-D	Within-person slope (fixed effect)	0.13	[−0.31, 0.51]	99.9	–	–	1.004	0
H1b-daily	M1-D	Between-person slope	−0.40	[−1.15, 0.29]	94.7	–	–	1.012	0
H1b-daily	M4-D	Between-person slope (fixed effect)	−0.40	[−1.11, 0.30]	95.2	–	–	1.004	0
H2a-daily	M2-D	SD of genre coefficients	0.63	[0.41, 0.87]	–	1.6	25.6	1.015	0
H2a-daily	M3-D	τ (hierarchical-prior scale)	0.22	[0.01, 0.57]	–	0.10	2.7	1.007	1
H2a-daily	M4-D	σ_s (random-slope SD)	0.49	[0.03, 1.30]	–	1.2	16.6	1.004	0
H2b-daily	M2-D	SD of genre coefficients	2.42	[1.49, 3.54]	–	9.3	75.9	1.015	0
H2b-daily	M3-D	τ (hierarchical-prior scale)	0.90	[0.06, 2.13]	–	3.5	36.6	1.007	1
H2b-daily	M4-D	σ_s (random-slope SD)	0.19	[0.01, 0.65]	–	0.14	2.7	1.004	0

Table A5.2. Exploratory daily-timescale hypothesis tests (see Daily Timescale in Results): posterior estimates, 95% credible intervals, posterior mass inside the ROPE for the total-playtime slopes (±1 slider point per hour; within-person rows are per hour of same-day play, between-person rows per hour of habitual daily play), descriptors of the genre spreads without a verdict (Genres > 1: expected number of the 14 genres more than one point from the average genre effect; Pairs > 1: share of genre pairs more than one point apart; posterior means, see Methods), and convergence diagnostics. M2-D rows are the SD across the 14 independently estimated genre coefficients, computed at each posterior draw, and include estimation error.

A6 Appendix F: Model Specifications and Priors

This appendix gives the equations and priors of the four biweekly models; the Methods describe what each model estimates and why we fitted it (Preregistered Models and Exploratory Models). The daily refits M1-D to M4-D use the same specifications with the daily outcome and exposures; `brms` then sets the intercept location and the standard-deviation scales from the life-satisfaction outcome (Methods, Daily Timescale).

A6.1 M1 specification: Total Playtime

M1, the preregistered model for H1, regresses SWEMWBS on the within-person and between-person parts of total playtime, with a random intercept and a random within-person playtime slope for each participant.

$$
\begin{aligned}
\text{SWEMWBS}_{ij} &\sim \text{Normal}\ (\mu_{ij}, \sigma) \\
\mu_{ij} &= (\alpha + u_{0j}) + (\beta_W + u_{1j})\,\text{Time}_{\text{within},ij} + \beta_B\,\text{Time}_{\text{between},j} \\
\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} &\sim \text{Normal}\ (\mathbf{0}, \boldsymbol{\Sigma}_u), \quad \boldsymbol{\Sigma}_u = \begin{pmatrix} \sigma_{u0}^2 & \rho_u\,\sigma_{u0}\,\sigma_{u1} \\ \rho_u\,\sigma_{u0}\,\sigma_{u1} & \sigma_{u1}^2 \end{pmatrix} \\
\alpha &\sim \text{Student-}\ t(3, \tilde{y}, 2.5) \\
\beta_W, \beta_B &\propto 1
\end{aligned}
\tag{A6.1}
$$

In this and the following equations, i indexes biweekly survey waves, j participants, and, from M2 onwards, g genres. $\text{Time}_{\text{within},ij}$ is participant j's average daily playtime on Nintendo, Xbox, and Steam (hours per day over the 14 days before wave i's scheduled survey) minus that participant's mean across all of their survey waves, and $\text{Time}_{\text{between},j}$ is that mean. α, β_W, and β_B are the population intercept and slopes; u_{0j} and u_{1j} are participant j's deviations from α and β_W, so that participant j's own intercept is $\alpha + u_{0j}$ and own within-person slope is $\beta_W + u_{1j}$; $\boldsymbol{\Sigma}_u$ is the covariance matrix of these deviations, with standard deviations σ_{u0} and σ_{u1} and correlation ρ_u; σ is the residual standard deviation; $\tilde{y}$ is the sample median of SWEMWBS; and $\propto 1$ denotes an improper flat prior.

A6.2 M2 specification: Independent Genre Coefficients

M2, the preregistered model for H2, replaces total playtime with the within-person and between-person exposures to each of the 14 genres, estimates their coefficients independently, and has a random intercept only.

$$
\begin{aligned}
\text{SWEMWBS}_{ij} &\sim \text{Normal}\ (\mu_{ij}, \sigma) \\
\mu_{ij} &= (\alpha + u_{0j}) + \sum_{g=1}^{G} \left(\beta_{W,g}\,\text{Time}_{\text{within},g,ij} + \beta_{B,g}\,\text{Time}_{\text{between},g,j} \right) \\
u_{0j} &\sim \text{Normal}\ (0, \sigma_{u0}) \\
\alpha &\sim \text{Student-}\ t(3, \tilde{y}, 2.5) \\
\beta_{W,g}, \beta_{B,g} &\propto 1
\end{aligned}
\tag{A6.2}
$$

Here g indexes the $G = 14$ genres. $\text{Time}_{\text{within},g,ij}$ and $\text{Time}_{\text{between},g,j}$ split participant j's exposure to genre g, the average daily hours spent on games carrying that genre label (a multi-genre game's full playtime counts towards each of its labels), in the same way as total playtime; $\beta_{W,g}$ and $\beta_{B,g}$ are their coefficients; and σ_{u0} is the standard deviation of the participant intercept deviations u_{0j}.

A6.3 M3 specification: Hierarchical Prior

M3, an exploratory model, keeps M2′s predictors and random intercept but draws the within-person and the between-person genre coefficients from two normal distributions whose centres and standard deviations it estimates.

$$
\begin{aligned}
\text{SWEMWBS}_{ij} &\sim \text{Normal}\ (\mu_{ij}, \sigma) \\
\mu_{ij} &= (\alpha + u_{0j}) + \sum_{g=1}^{G} \left(\beta_{W,g}\,\text{Time}_{\text{within},g,ij} + \beta_{B,g}\,\text{Time}_{\text{between},g,j} \right) \\
u_{0j} &\sim \text{Normal}\ (0, \sigma_{u0}) \\
\beta_{W,g} &\sim \text{Normal}\ (\gamma_W, \tau_{\text{within}}) \\
\beta_{B,g} &\sim \text{Normal}\ (\gamma_B, \tau_{\text{between}}) \\
\gamma_W, \gamma_B &\propto 1 \\
\tau_{\text{within}}, \tau_{\text{between}} &\sim \text{Half-Student-}\ t(3, 0, 2.5) \\
\alpha &\sim \text{Student-}\ t(3, \tilde{y}, 2.5)
\end{aligned}
\tag{A6.3}
$$

The symbols are as in M2. γ_W and γ_B are the centres and τ_{within} and τ_{between} the standard deviations of the distributions from which the within-person and between-person genre coefficients are drawn. γ_W and γ_B receive the same improper flat prior as the slopes in M1 and M2, and τ_{within} and τ_{between} describe the size of the genre differences in the within-person (H2a) and between-person (H2b) dimensions. Because the centred form of this model, in which the sampler draws each genre coefficient directly from its normal distribution, produced dozens of divergent transitions in earlier fits, we estimate it in the equivalent non-centred form $\beta_{W,g} = \gamma_W + \tau_{\text{within}}\,z_{W,g}$ and $\beta_{B,g} = \gamma_B + \tau_{\text{between}}\,z_{B,g}$, with

$z_{W,g}, z_{B,g} \sim \text{Normal}\ (0, 1)$. In `brms` this is a model with slopes γ_W and γ_B on the summed genre exposures plus two genre-level random terms weighted by the genre exposures (the multi-membership construction that M4 also uses); it changes neither the likelihood nor the priors on γ and τ.

A6.4 M4 specification: Multi-Membership

M4, an exploratory model, keeps M1′s total-playtime terms and participant effects and lets the genre composition of each wave adjust the within-person and between-person playtime slopes. Formally, this is a multi-membership structure: every wave belongs to all 14 genres, and only the membership weights vary, each equal to the genre's share of the wave's summed genre exposure.

$$
\begin{aligned}
\text{SWEMWBS}_{ij} &\sim \text{Normal}\ (\mu_{ij}, \sigma) \\
\mu_{ij} &= (\alpha + u_{0j}) + \Big(\beta_W + u_{1j} + \sum_{g=1}^{G} w_{g,ij}\, s_{W,g}\Big)\, \text{Time}_{\text{within},ij} \\
&\quad + \Big(\beta_B + \sum_{g=1}^{G} w_{g,ij}\, s_{B,g}\Big)\, \text{Time}_{\text{between},j} \\
\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} &\sim \text{Normal}\ (\mathbf{0}, \boldsymbol{\Sigma}_u) \\
\begin{pmatrix} s_{W,g} \\ s_{B,g} \end{pmatrix} &\sim \text{Normal}\ (\mathbf{0}, \boldsymbol{\Sigma}_s) \\
\boldsymbol{\Sigma}_s &= \begin{pmatrix} \sigma_{s,W}^2 & \rho_s\, \sigma_{s,W}\, \sigma_{s,B} \\ \rho_s\, \sigma_{s,W}\, \sigma_{s,B} & \sigma_{s,B}^2 \end{pmatrix} \\
\alpha &\sim \text{Student-}\ t(3, \tilde{y}, 2.5) \\
\beta_W, \beta_B &\propto 1 \\
\sigma_{s,\ \text{within}}, \sigma_{s,\ \text{between}} &\sim \text{Half-Student-}\ t(3, 0, 2.5) \\
\begin{pmatrix} 1 & \rho_s \\ \rho_s & 1 \end{pmatrix} &\sim \text{LKJ}\ (1)
\end{aligned}
\tag{A6.4}
$$

The participant terms are as in M1. $w_{g,ij}$ is genre g's exposure hours in participant j's wave i divided by the summed exposure hours of all G genres in that wave, so a wave's weights sum to one; because exposure uses full attribution, a game with two genre labels adds its hours to both. Waves without genre-labelled play, most of them without any recorded play, receive $w_{g,ij} = 1/G$. $s_{W,g}$ and $s_{B,g}$ are genre g's adjustments to the within-person and between-person playtime slopes; because the weights describe each wave's genre mix, the between-person adjustment follows the wave's mix rather than the participant's habitual mix. The two adjustments are drawn jointly, with standard deviations $\sigma_{s,\ \text{within}}$ and $\sigma_{s,\ \text{between}}$ (written $\sigma_{s,W}$ and $\sigma_{s,B}$ in the equation) and correlation ρ_s; the LKJ(1) prior on their 2 × 2 correlation matrix is uniform over ρ_s.

Because this use of multi-membership is unconventional, we validated the specification by parameter recovery. The simulation fits the manuscript's own M4 code, read from this document at run time, to 20 synthetic datasets of 200 participants over six waves with known within-person and between-person genre heterogeneity. The 95% credible intervals covered each of the ten true values in at least 19 of the 20 replications, with no divergent transitions. At this sample size, a fraction of the real one, the genre-level standard deviations were estimated with upward bias (posterior means of 0.12 within and 0.06 between persons for true values of 0.04), and the correlation between the within-person and between-person adjustments was weakly identified (true value 0.3, averaged 95% CI [−0.93, 0.93]), so σ_s should be read with its interval rather than its point estimate (simulation report in the supplementary materials).

A6.5 Priors

All models use the `brms` default priors (Bürkner, 2017). The slopes have improper flat priors, so that the data alone determine their estimates, which are directly comparable with the preregistered maximum-likelihood analysis; the exception is M3, whose genre coefficients receive the hierarchical prior that defines the model. `brms` places the intercept prior on the intercept with all predictors at their sample means, that is, on expected wellbeing at average playtime (for M3, at the mean summed genre exposure), from which `brms` recovers α after sampling; this prior is Student-t(3, 3.3, 2.5), centred at the sample median of SWEMWBS. The residual standard deviation σ and every other standard deviation, including the participant-level standard deviations and the genre-level spreads τ (M3) and σ_s (M4), have half-Student-t(3, 0, 2.5) priors; their scale is the larger of 2.5 and the median absolute deviation of the outcome, which for SWEMWBS gives 2.5. Both τ and σ_s are standard deviations of `brms` group-level terms (for τ, in the non-centred form of M3 described above), so `brms` assigns this prior to them by default. Each correlation matrix has an LKJ(1) prior, which for a 2 × 2 matrix is uniform over the single correlation. An earlier version used weakly informative Student-t priors on the slopes and intercept, a Half-Normal(0, 0.1) hyperprior on τ and a Normal(0, 0.3) prior on σ_s; following co-author review, and after those results were known, we adopted the defaults so that no analyst-chosen scale enters the models.